

Tutorial

Artificial Intelligence in Microfluidic Circuits

Speakers

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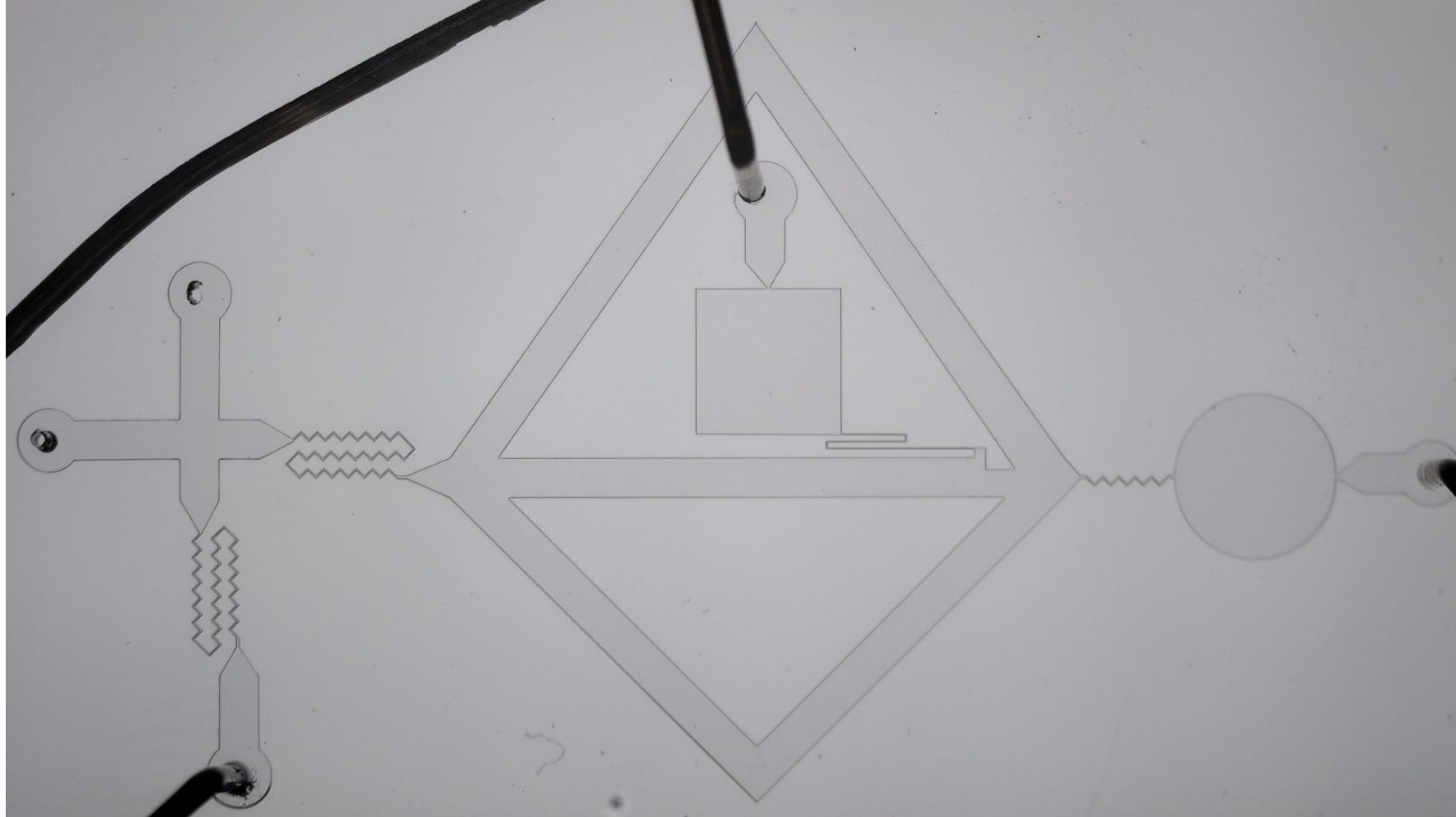


DSP in Microfluidic MC Systems

Summary

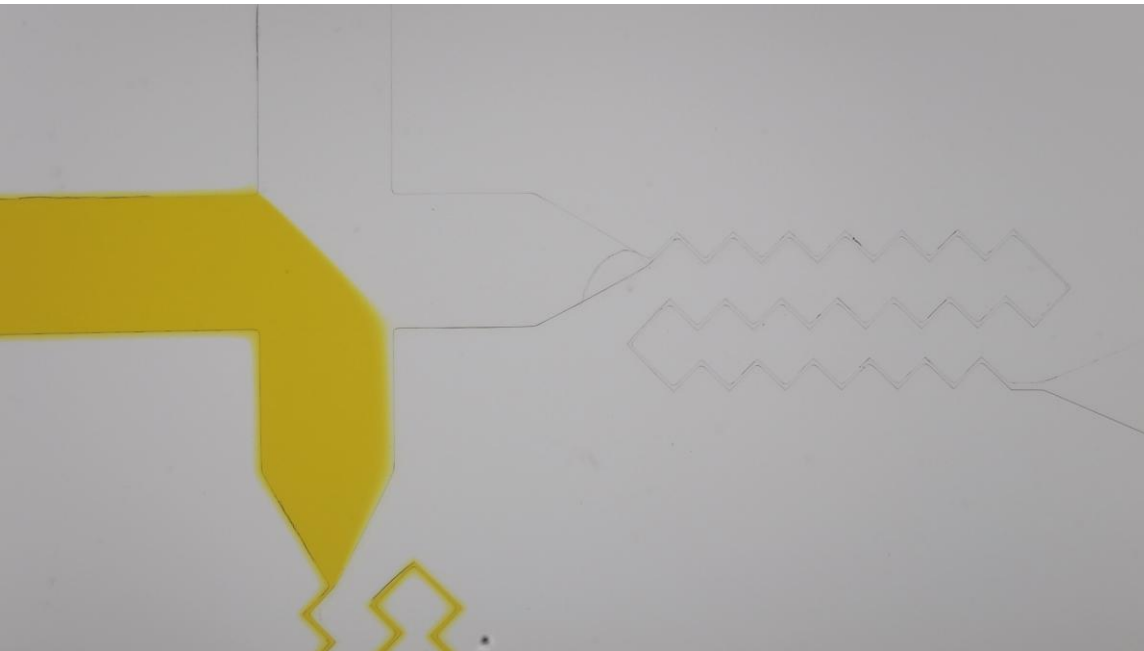
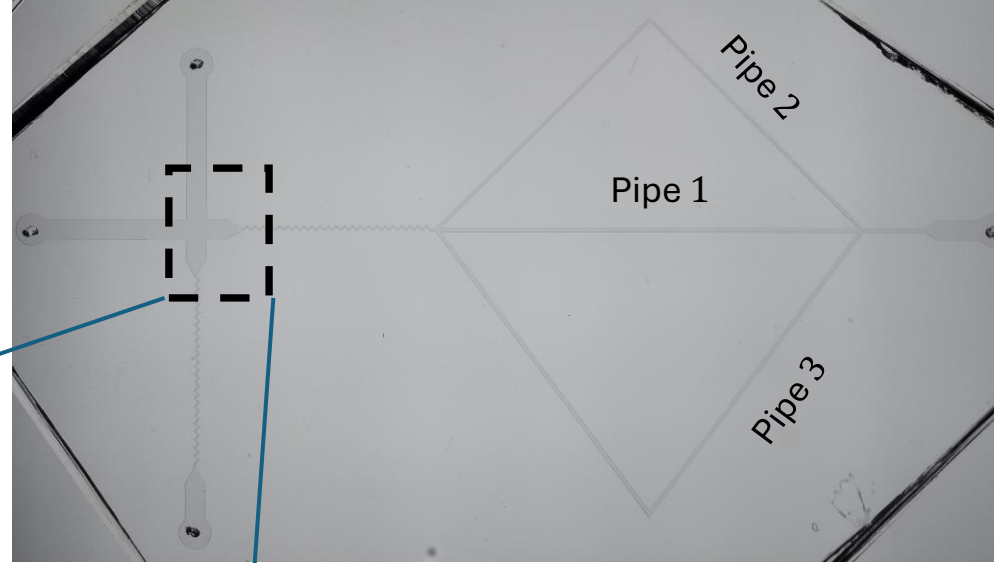
- Introduction
- Analysis of microfluidic flows
- Synthesis of microfluidic designs in COMSOL

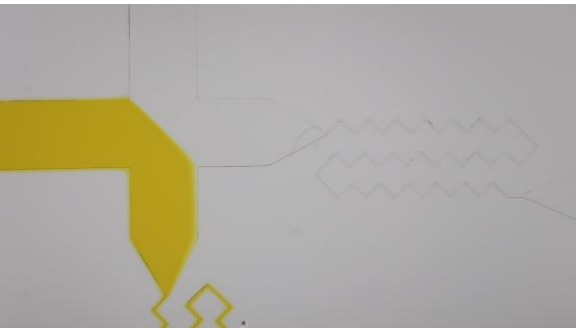
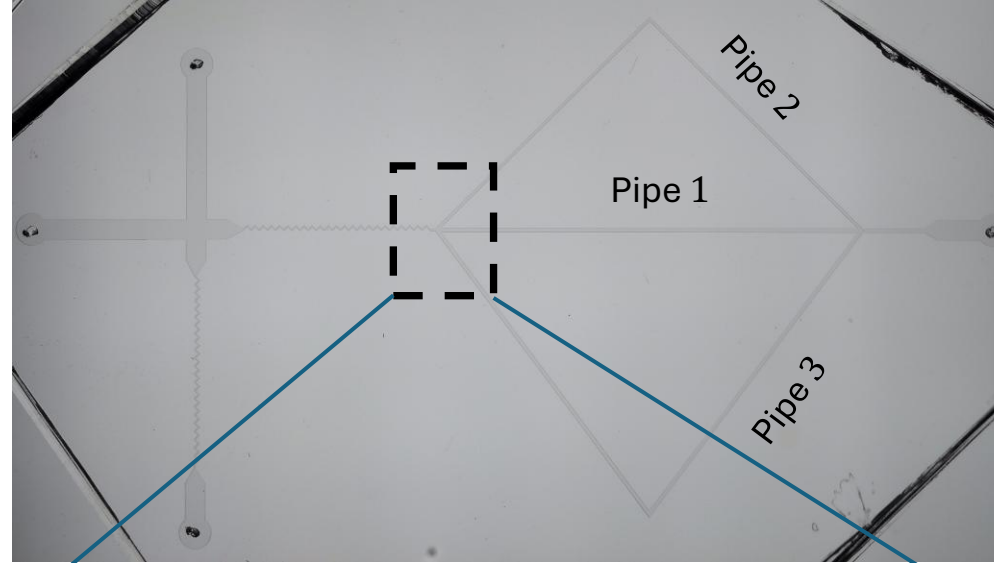
Microfluidic Circuit

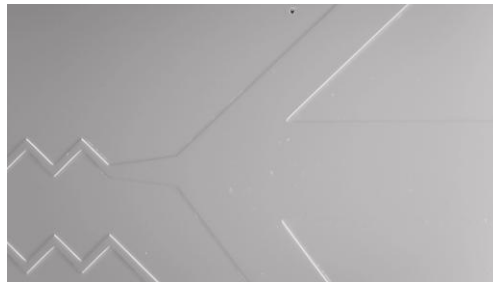
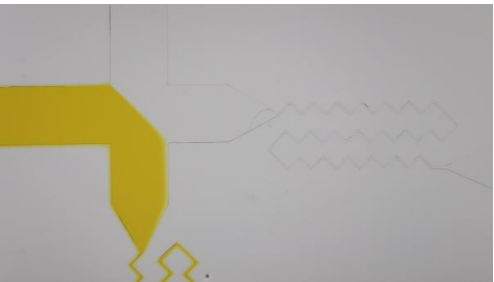
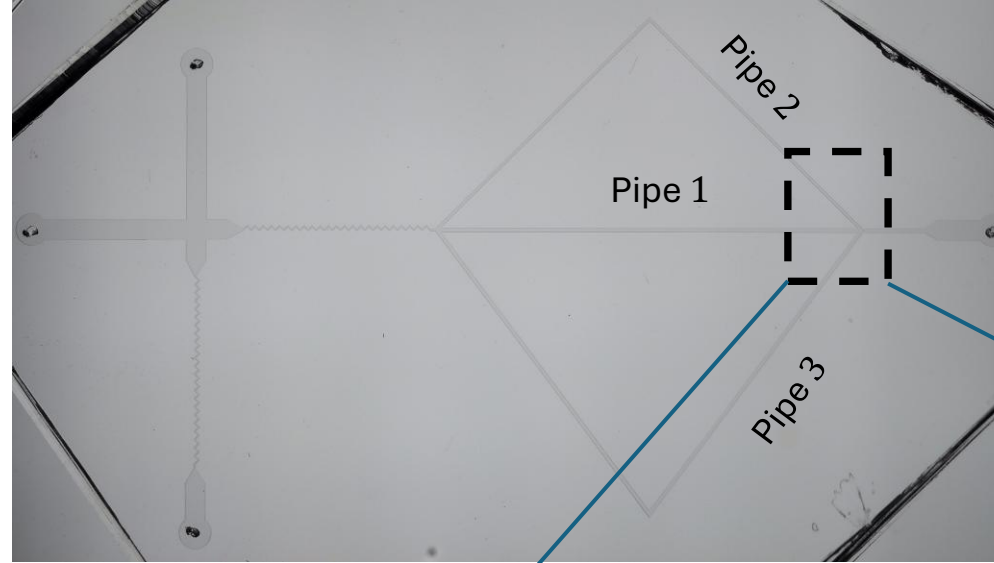


19 cm

35 cm

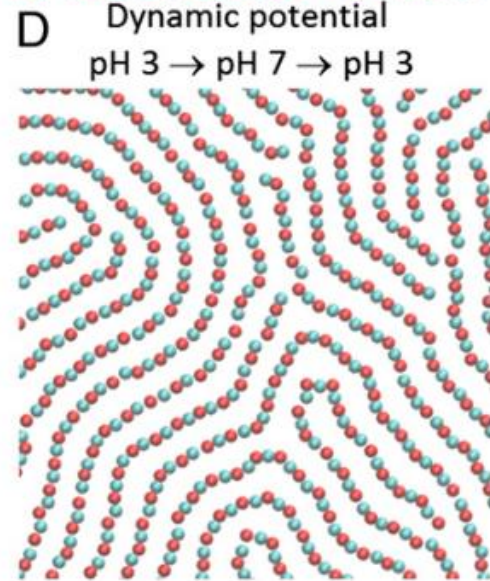
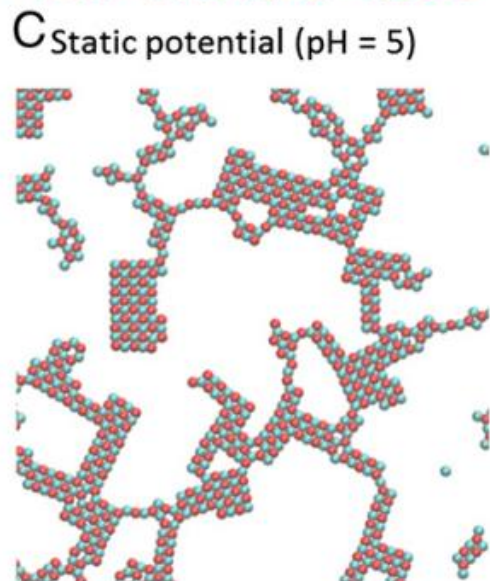
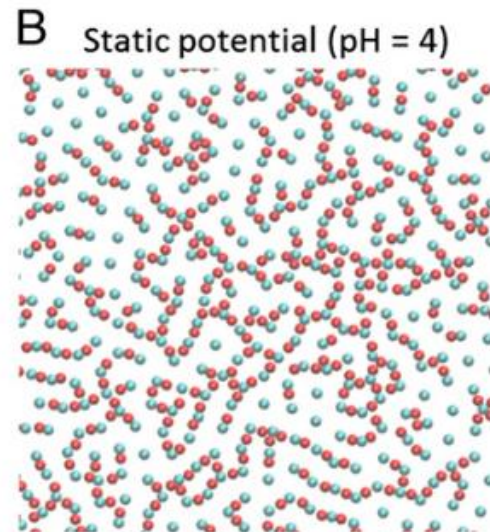
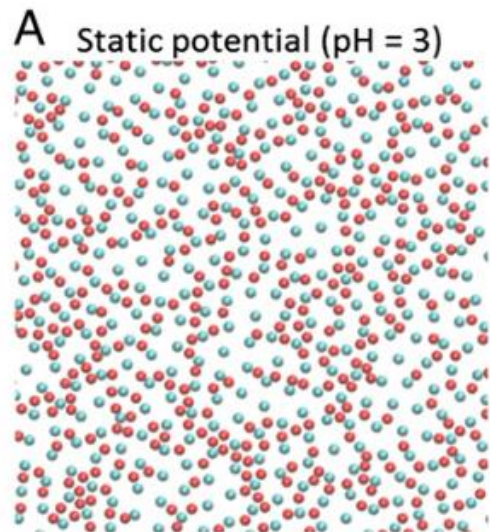






Volumetric Flow: $Q < 20 \mu\text{L}/\text{min}$

Traveling time: seconds – minutes



Chemical pH Oscillators

Self-Assembly

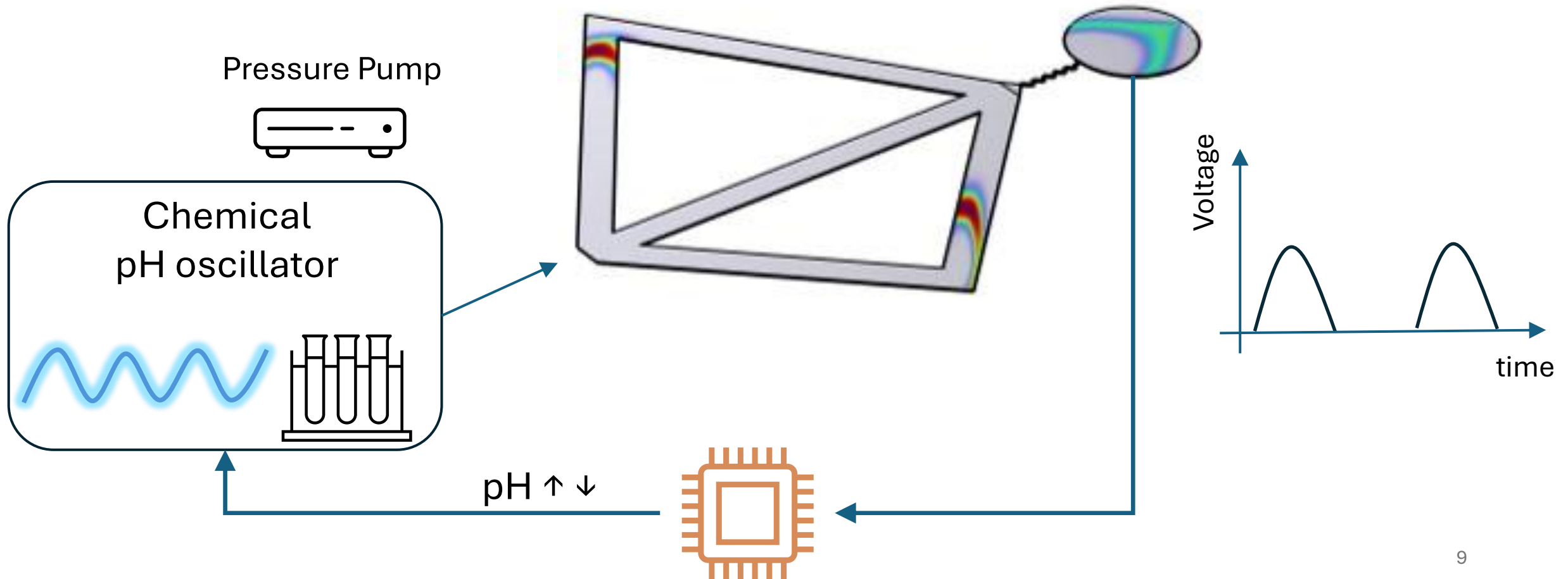
Periodic drug delivery

Molecular Motors

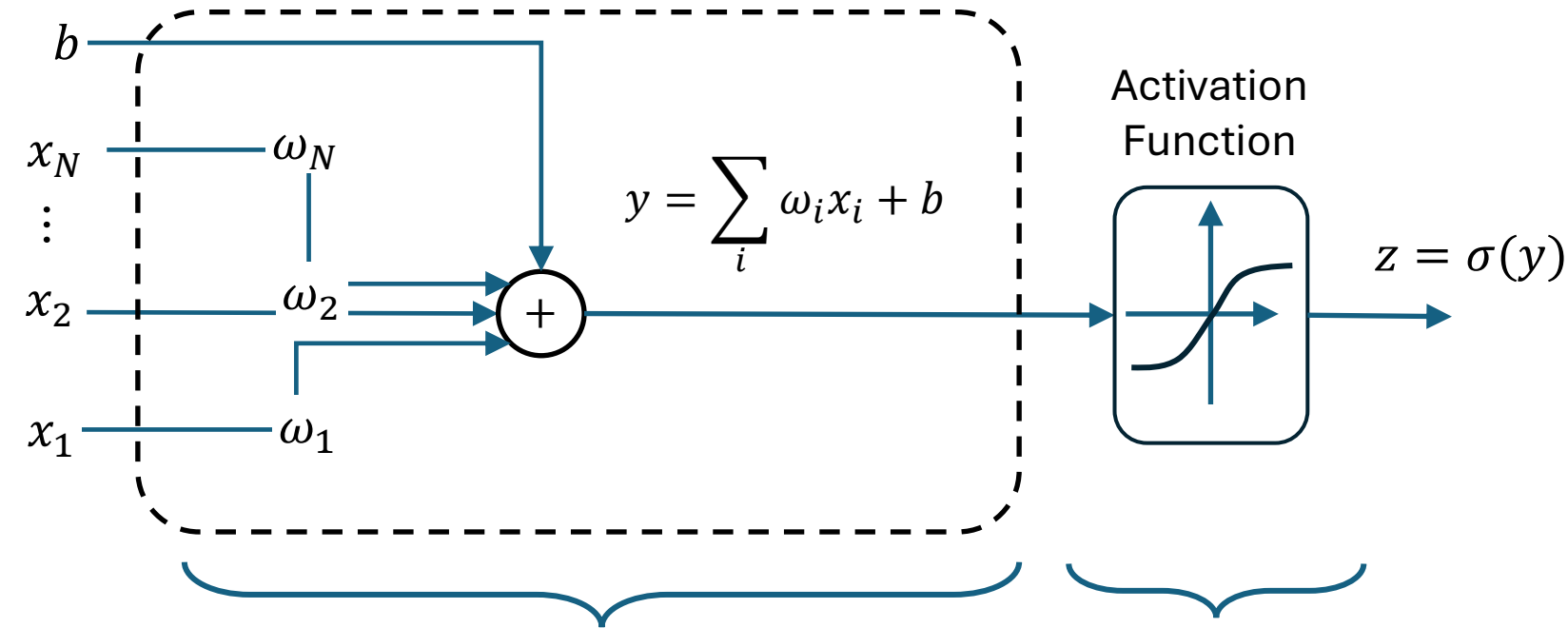
Fast-tracking Applications

- M. Orbán, K. Kurin-Csörgei, and I. R. Epstein, “pH-Regulated Chemical Oscillators,” *Acc. Chem. Res.*, vol. 48, no. 3, pp. 593–601, Mar. 2015, doi: [10.1021/ar5004237](https://doi.org/10.1021/ar5004237).

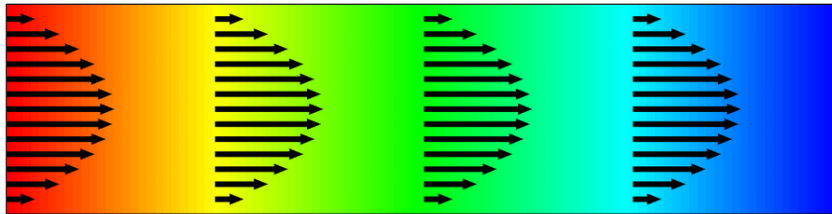
Application context



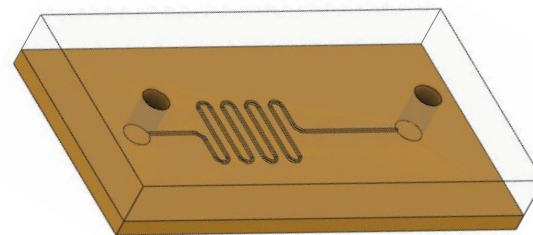
Single Neuron Scheme



Transport of Diluted Species



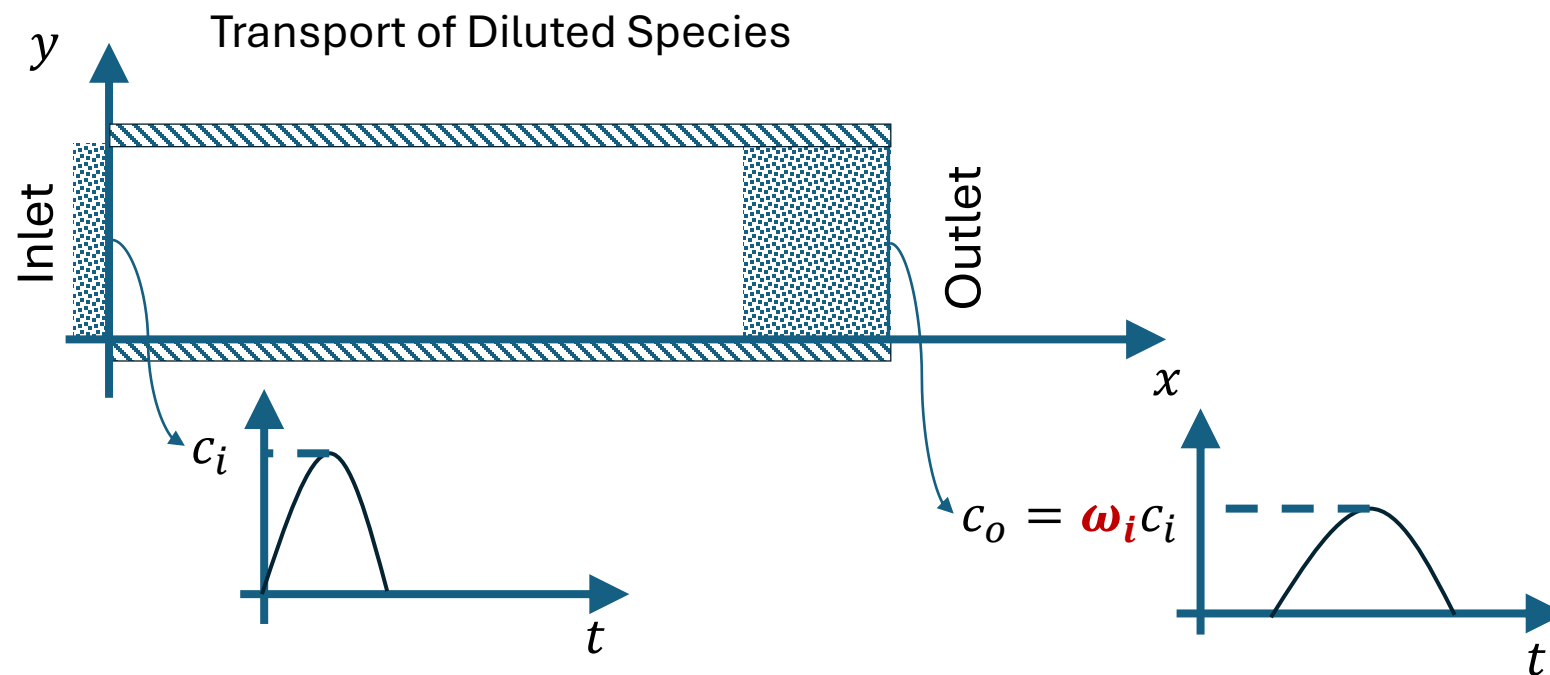
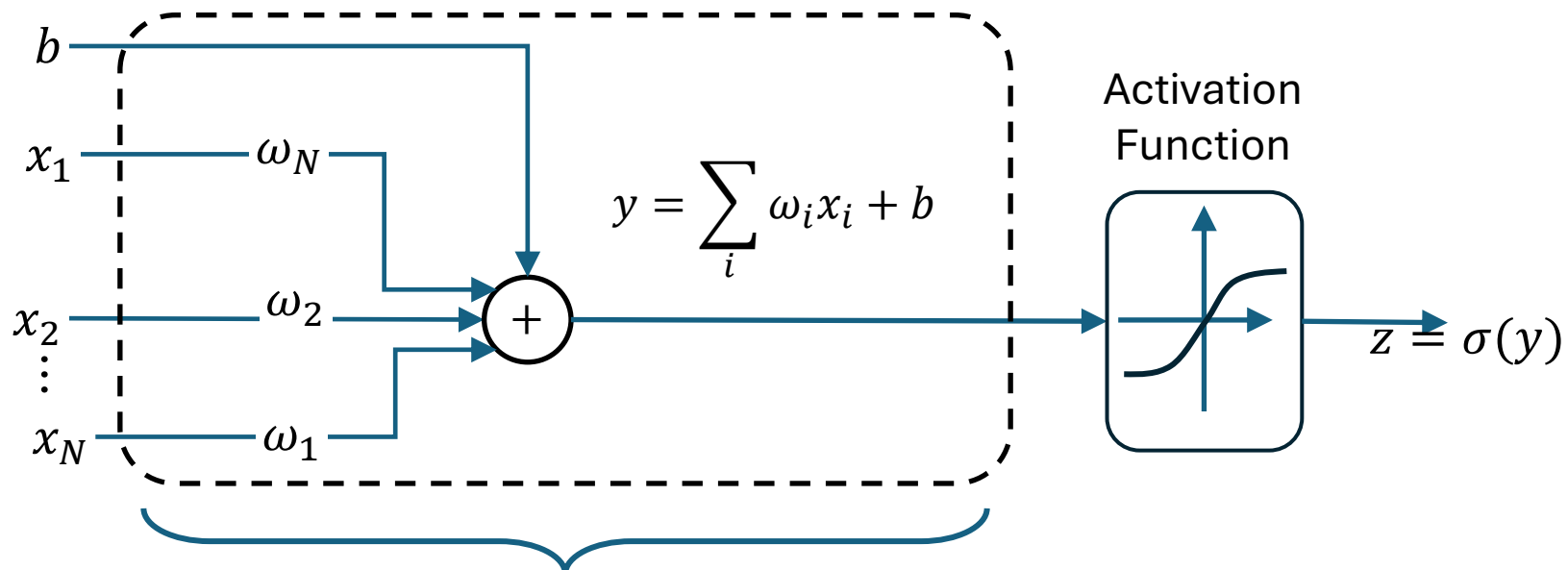
Chemical Reaction



Microfluidic Circuit

[1] A. O. Bicen and I. F. Akyildiz, "System-Theoretic Analysis and Least-Squares Design of Microfluidic Channels for Flow-Induced Molecular Communication," IEEE Transactions on Signal Processing, vol. 61, no. 20, pp. 5000–5013, Oct. 2013, doi: 10.1109/TSP.2013.2274959.

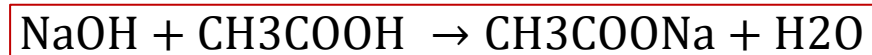
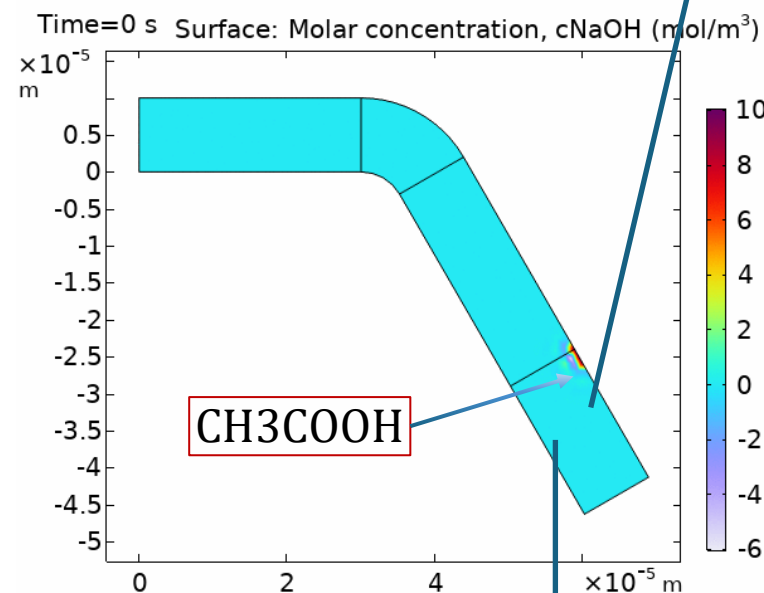
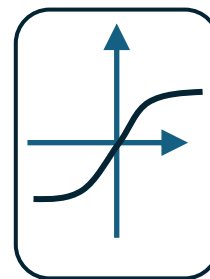
Single Neuron Scheme



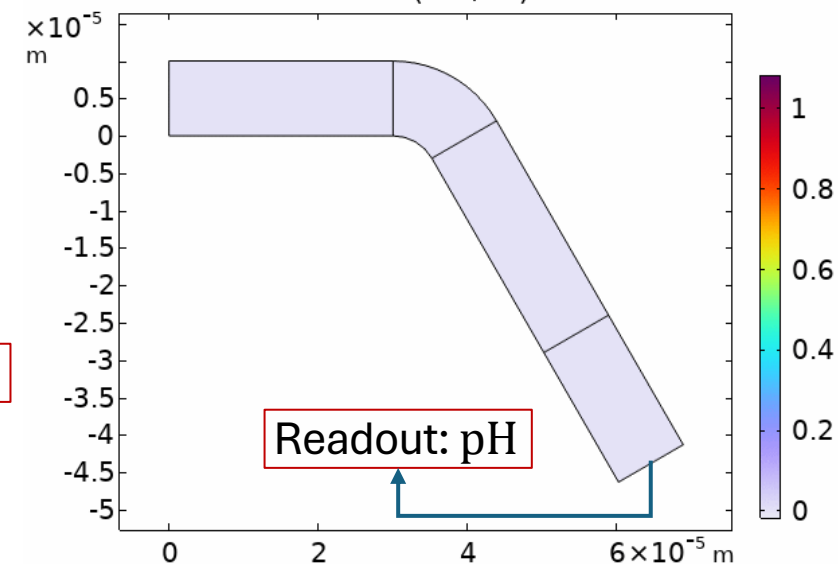
Concept



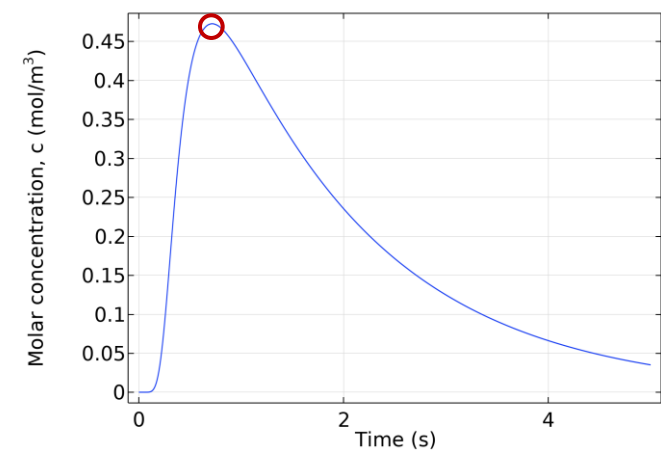
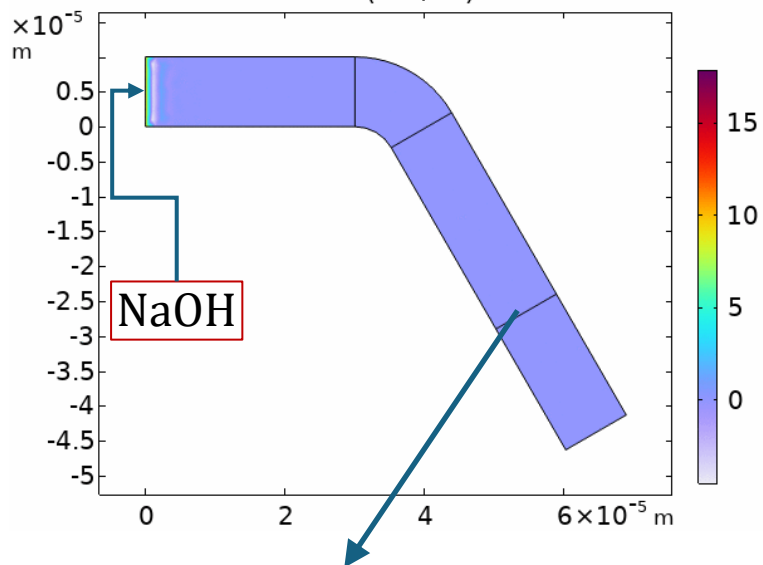
Activation
Function



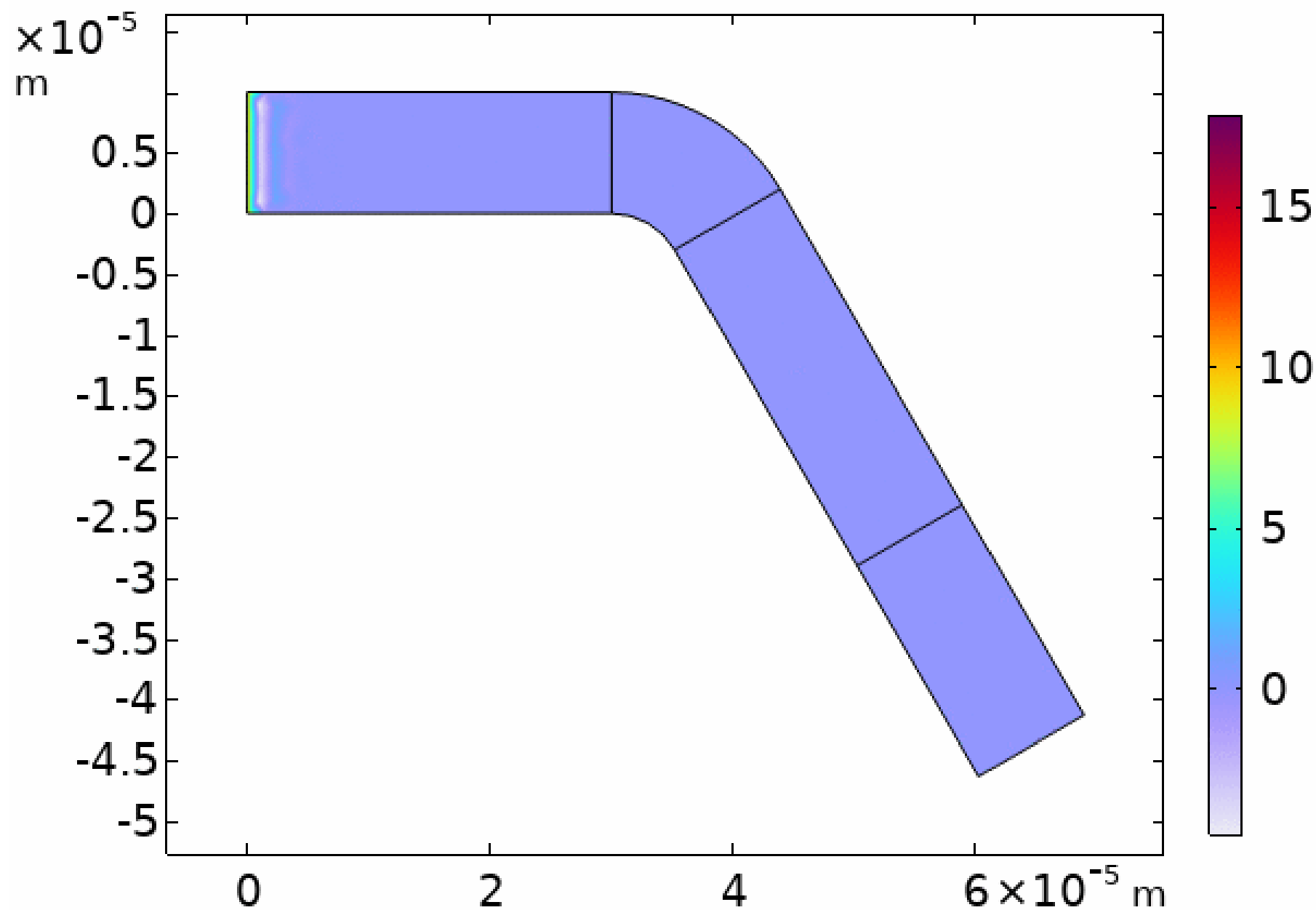
Time=0 s Surface: Molar concentration, $c_{\text{CH}_3\text{COONa}}$ (mol/m^3)



Time=0 s Surface: Molar concentration, $c_{\text{CH}_3\text{COOH}}$ (mol/m^3)

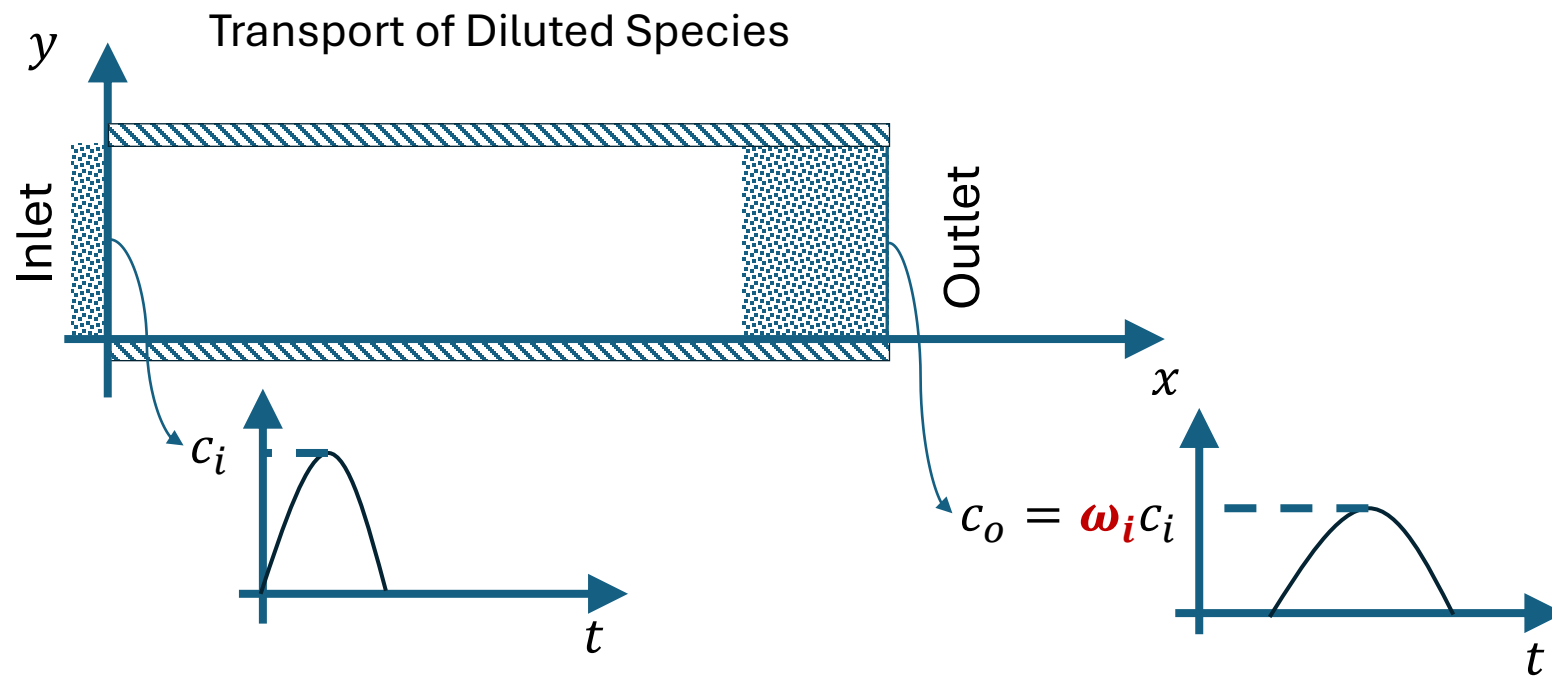


Time=0 s Surface: Molar concentration, $c_{\text{CH}_3\text{COOH}}$
(mol/m³)



Summary

- Introduction
- Analysis of microfluidic flows
- Synthesis of microfluidic pipes in COMSOL



Analysis

Derivative of the Step-response

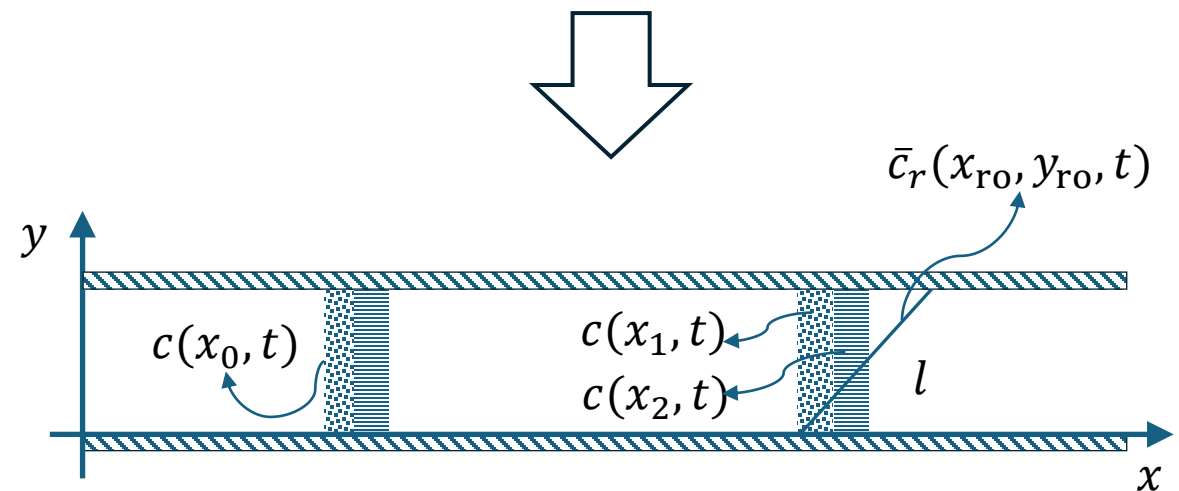
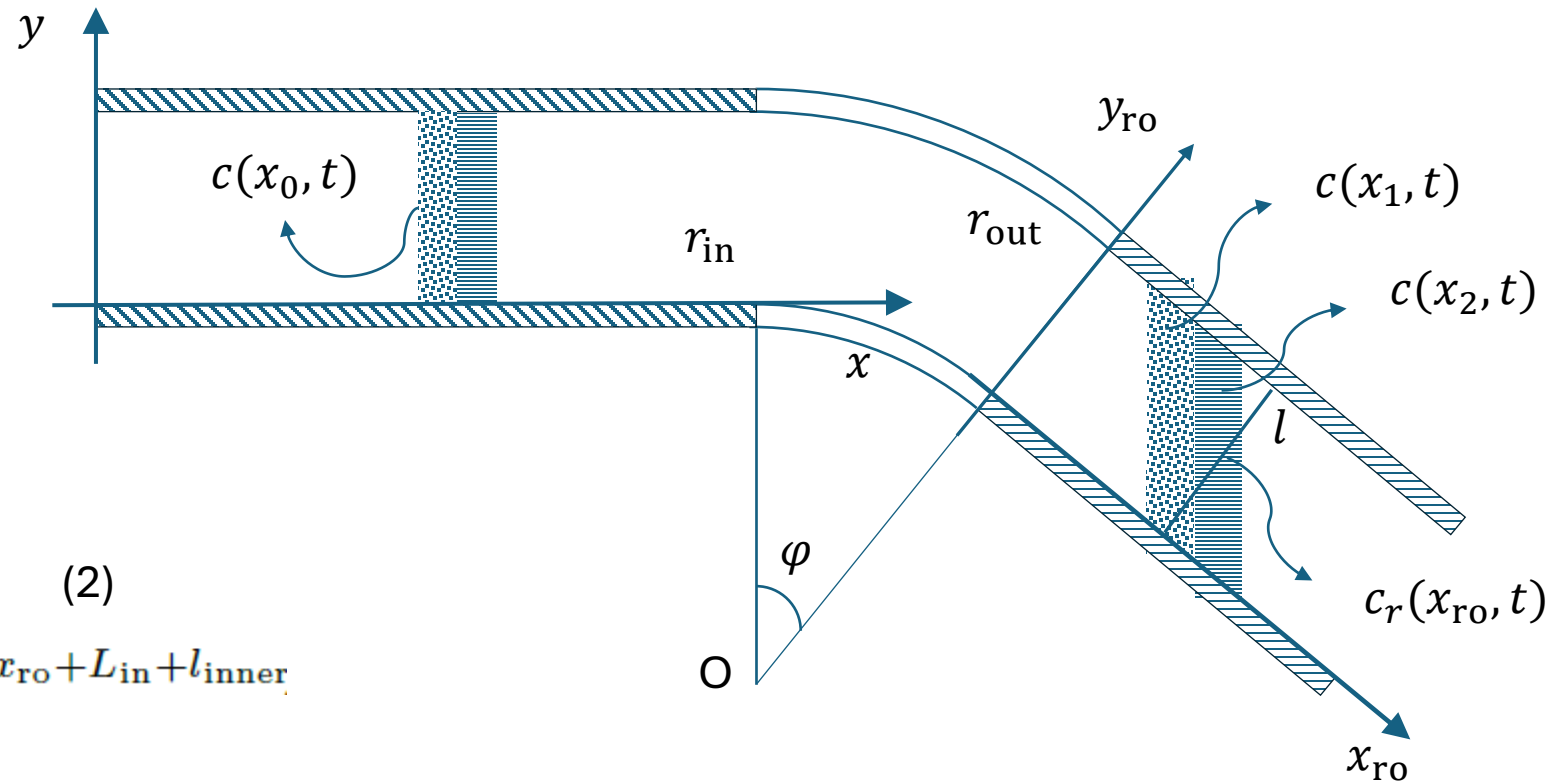
$$h(t) = \left. \frac{\partial}{\partial t} \bar{c}_r(x_{ro}, t) \right|_{C_0=1} \quad (1)$$

$$\bar{c}_r(x_{ro}, t) = \frac{1}{2r} \int_0^{2r} c(x + \varphi y_{ro}, t) dy_{ro} \Big|_{x=x_{ro}+L_{in}+l_{inner}} \quad (2)$$

$$c(x, t) = \frac{1}{2} C_0 \operatorname{erfc} \left(\frac{x - v_{\text{char}} t}{2\sqrt{D_{\text{eff}} t}} \right) \quad (3)$$

$$v_{\text{char}} = \frac{r^2 \Delta p}{3\mu L} \quad (4) \quad D_{\text{eff}} = \frac{r^2 v_{\text{char}}^2}{48D} \quad (5)$$

$$\Delta p = \frac{8\mu L}{\pi r^4} Q \quad (6)$$

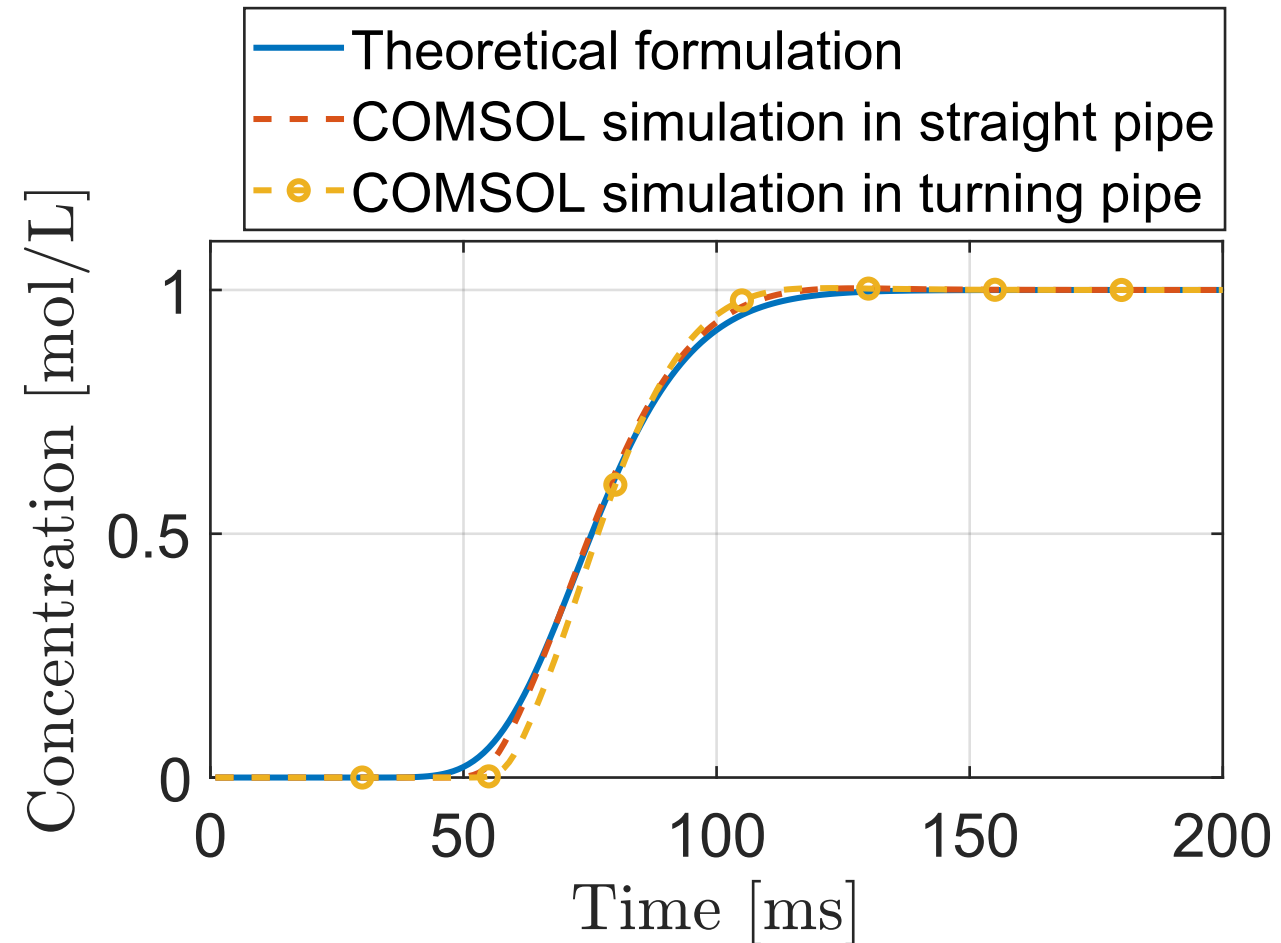
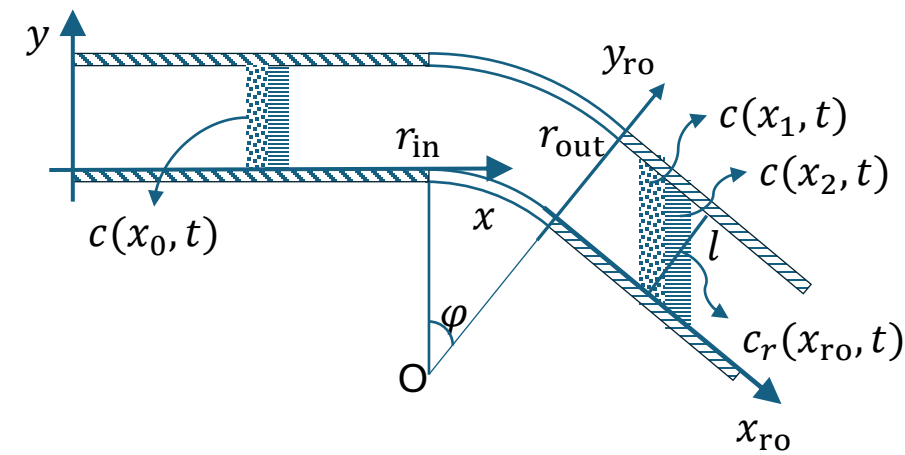


Analysis

Derivative of the Step-response

$$h(t) = \left. \frac{\partial}{\partial t} \bar{c}_r(x_{ro}, t) \right|_{C_0=1} \quad (1)$$

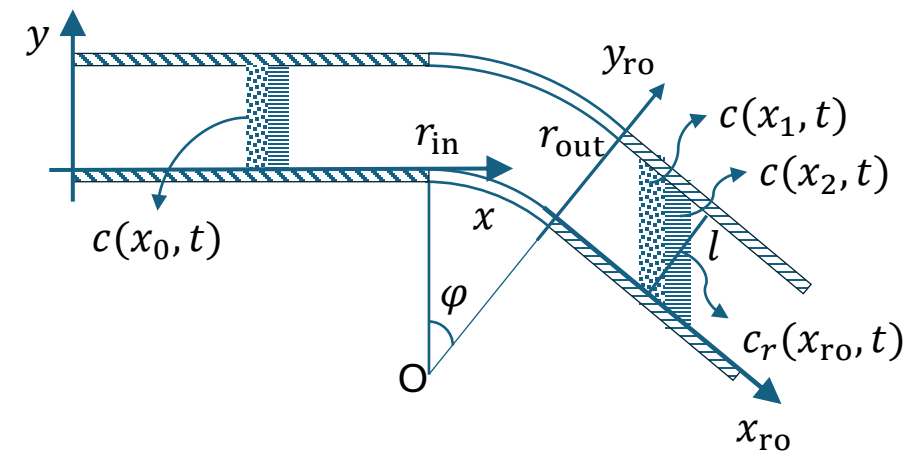
$$\bar{c}_r(x_{ro}, t) = \frac{C_0}{4r} \left[\operatorname{erfc} \left(\frac{x - tv + \varphi y}{2 \sqrt{D_{\text{eff}} t}} \right) (x - tv + \varphi y) - 2 \sqrt{\frac{D_{\text{eff}} t}{\pi}} e^{-\frac{(x - tv + \varphi y)^2}{4 D_{\text{eff}} t}} \right] \bigg|_{y=0}^{y=2r} \quad (2)$$



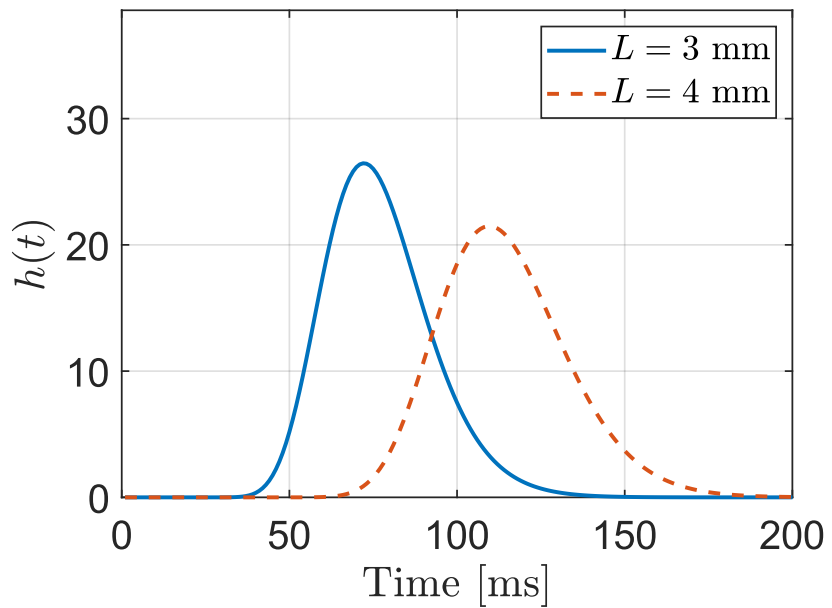
Analysis

Derivative of the Step-response

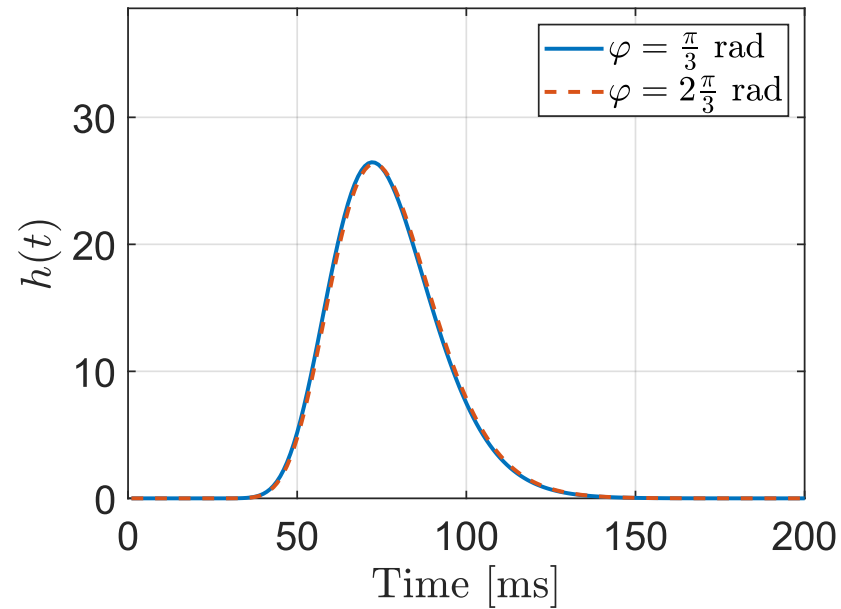
$$h(t) = \left. \frac{\partial}{\partial t} \bar{c}_r(x_{ro}, t) \right|_{C_0=1} \quad (1)$$



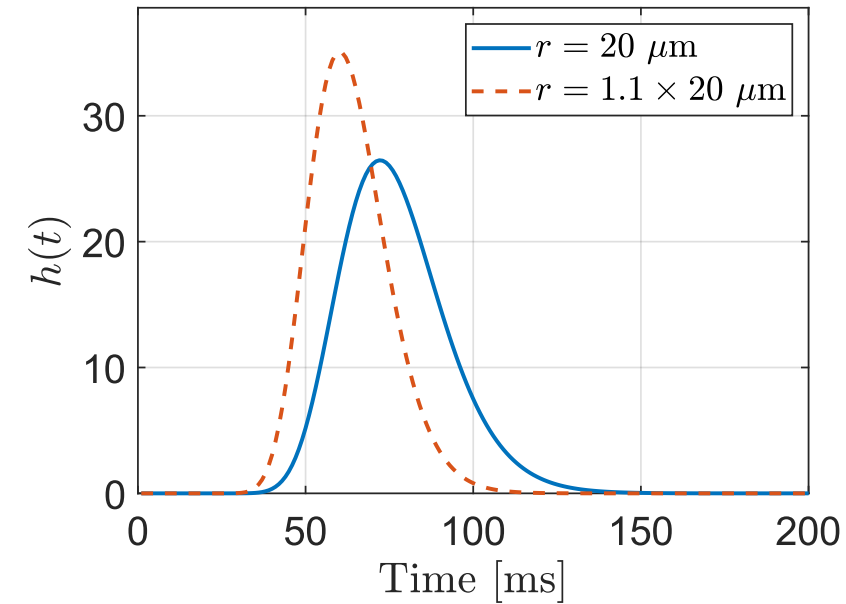
$$\varphi = \frac{\pi}{3} \quad r = 20 \mu\text{m}$$



$$L = 3 \text{ mm} \quad r = 20 \mu\text{m}$$



$$L = 3 \text{ mm} \quad \varphi = \frac{\pi}{3}$$

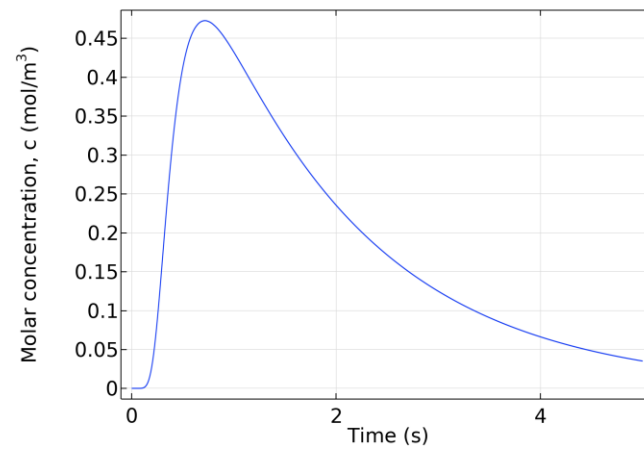
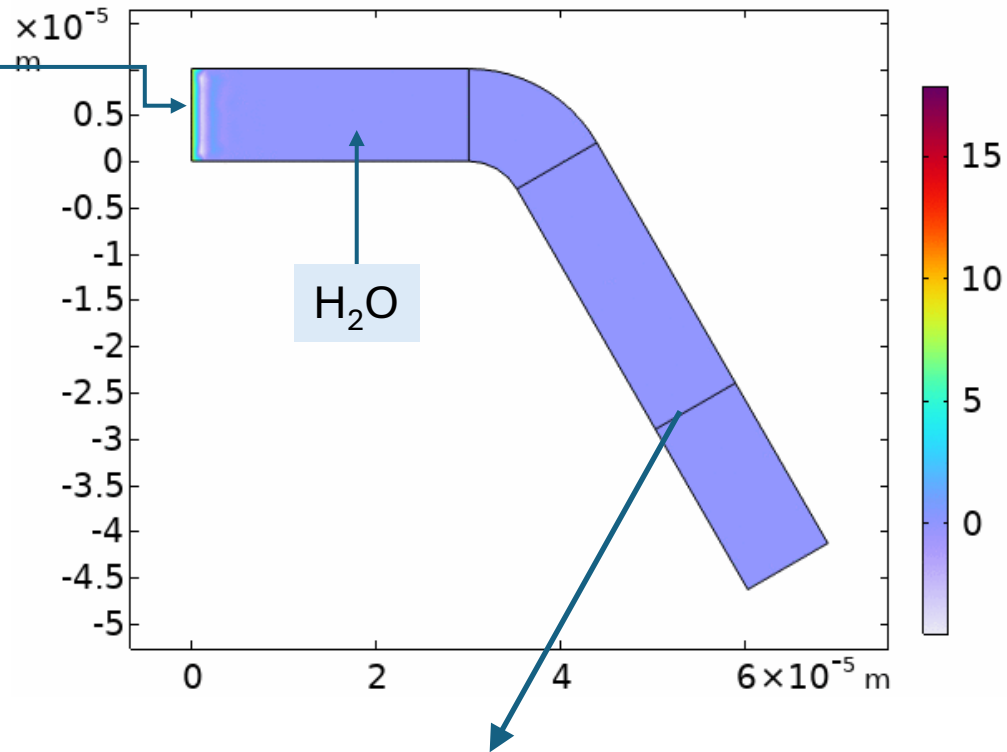


Summary

- Introduction
- Analytic evaluation of advection and diffusion in microfluidic channels
- Synthesis of microfluidic pipes in COMSOL

NaOH

Time=0 s Surface: Molar concentration, $c_{\text{CH}_3\text{COOH}}$
(mol/m³)



File

New

Model Wizard

Blank Model

File

Select Space Dimension

3D

2D Axisymmetric

2D

File

Select Physics

Type filter text

Recently Used

Laminar Flow (spf)

Transport of Diluted Species (tds)

Reaction Engineering (re)

> AC/DC

> Acoustics

> Chemical Species Transport

> Electric Discharge

> Electrochemistry

> Fluid Flow

> Heat Transfer

> Optics

> Plasma

> Radio Frequency

> Semiconductor

> Structural Mechanics

> Mathematics

Add

Added physics interfaces:

Laminar Flow (spf)

Transport of Diluted Species (tds)

Review Physics Interface

Laminar Flow (spf)

Dependent Variables

Velocity field (m/s):

u

Velocity field components:

u

v

w

Pressure (Pa):

p

Select Physics

Type filter text

Recently Used

-  Laminar Flow (spf)
-  Transport of Diluted Species (tds)
-  Reaction Engineering (re)
- >  AC/DC
- >  Acoustics
- >  Chemical Species Transport
- >  Electric Discharge
- >  Electrochemistry
- >  Fluid Flow
- >  Heat Transfer
- >  Optics
- >  Plasma
- >  Radio Frequency
- >  Semiconductor
- >  Structural Mechanics
- >  Mathematics

Laminar Flow

Add

Added physics interfaces:

-  Laminar Flow (spf)
-  Transport of Diluted Species (tds)

Review Physics Interface

Laminar Flow (spf)

Dependent Variables

Velocity field (m/s):

Velocity field components:

Pressure (Pa):

Fluid Velocity Components

(u, v, w) [m/s]

Inlet Pressure

p [Pa]

File

Untitled.mph - COMSOL Multiphysics

Select Physics

Type filter text

Recently Used

Laminar Flow (spf)

Transport of Diluted Species (tds)

Reaction Engineering (re)

> AC/DC

> Acoustics

> Chemical Species Transport

> Electric Discharge

> Electrochemistry

> Fluid Flow

> Heat Transfer

> Optics

> Plasma

> Radio Frequency

> Semiconductor

> Structural Mechanics

> Mathematics

Add

Added physics interfaces:

Laminar Flow (spf)

Transport of Diluted Species (tds)

Review Physics Interface

Transport of Diluted Species (tds)

Dependent Variables

Number of species: 1

Concentrations (mol/m³): c

+

Transport of Diluted Species

Concentration of Species
 $c \text{ [mol/m}^3\text{]}$

23

Select Study

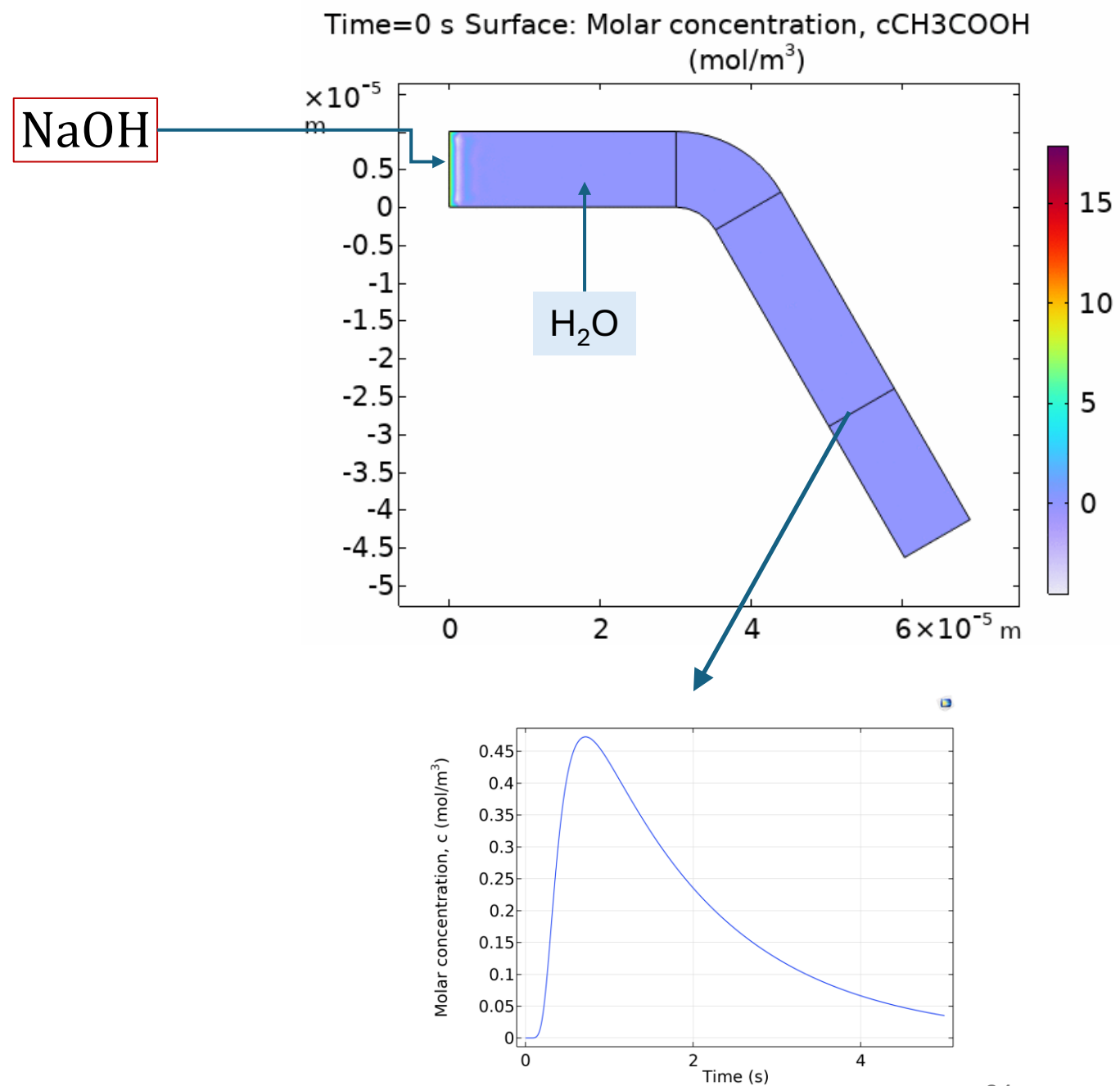
- General Studies
 - Stationary
 - Time Dependent
 - Empty Study

Stationary:

H₂O Flow

Time Dependent:

Transport of Diluted Species NaOH



File Home Definitions Geometry Materials Physics Mesh Study Results Developer

Application Builder Model Manager Component 1 Add Component Parameters Variables Functions Parameters Case Build All Import LiveLink Part Libraries Add Material Laminar Flow Add Physics Add Mathematics Build Mesh Mesh 1 Compute Study 1 Add Study Select Plot Group Add Plot Group Result Templates Windows Reset Desktop

Workspace Model Definitions Geometry Materials Physics Mesh Study Results Layout

Model Builder

Type filter text

- Untitled.mph (root)
 - Global Definitions
 - Parameters 1
 - Materials
 - Component 1 (comp 1)
 - Definitions
 - Geometry 1**
 - Materials
 - Laminar Flow (spf)
 - Fluid Properties 1
 - Initial Values 1
 - Wall 1
 - Transport of Diluted Species (tds)
 - Fluid 1
 - No Flux 1
 - Initial Values 1
 - Multiphysics
 - Mesh 1
 - Study 1
 - Step 1: Stationary
 - Results

Settings

Geometry

Build All

Label: Geometry 1

Units

☐ Scale values when changing units

Length unit: m

Angular unit: Degrees

Advanced

Geometry representation: CAD kernel

☒ Design Module Boolean operations

Default repair tolerance: Automatic

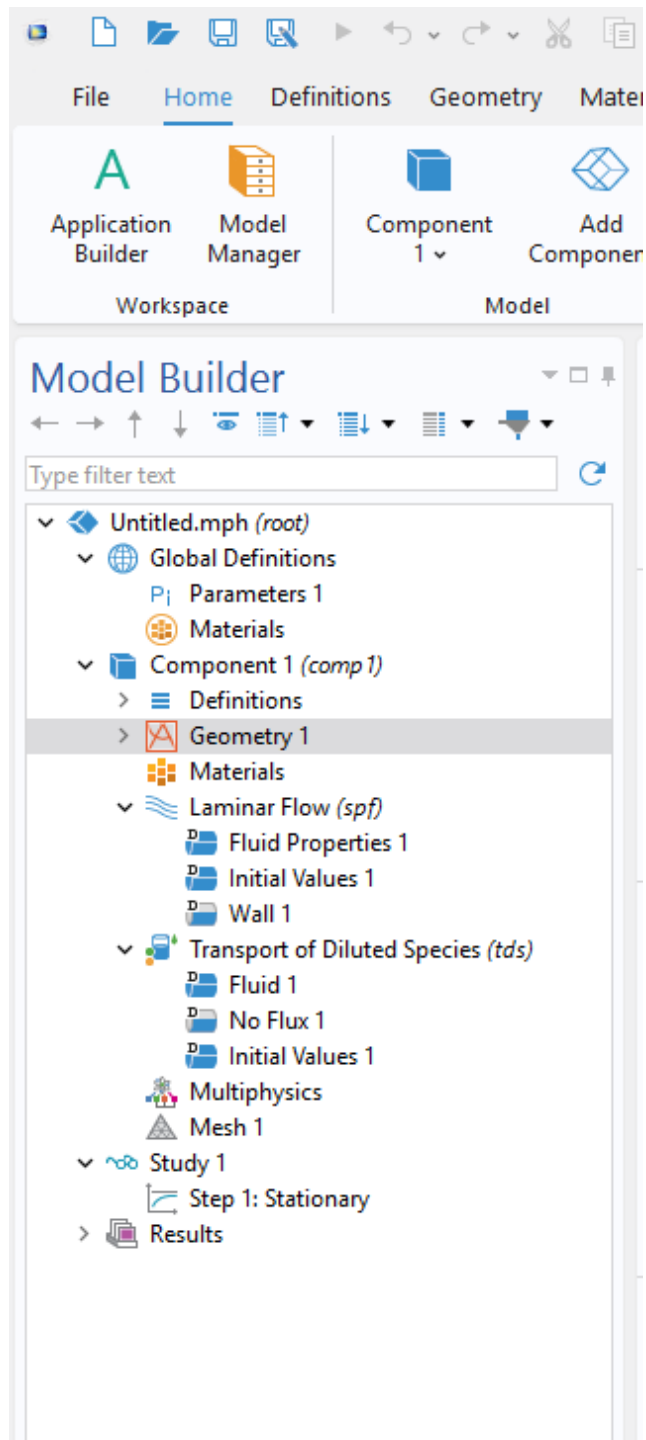
☒ Build new operations automatically

☒ Build automatically when leaving geometry

Cleanup

☒ Automatic detection of small details

Graphics



Components:

- Geometry
- Laminar Flow Physics
- Transport of Diluted Species Physics
- Multiphysics
- Study
- Results

File Home Definitions **Geometry** Materials Physics Mesh Study Results Developer

Build All Import Defeaturing and Repair Cleanup Wizard Virtual Operations Cleanup Block Sphere Cone Cylinder More Primitives Select Work Plane Work Plane Extrude Revolve Sweep Loft Booleans and Partitions Transforms Extract Delete Part Libraries Create Part Load Part Selections Selection List Colors Programming Measure Measurements Clear Sequence

Model Builder

Type filter text

- Untitled.mph (root)
 - Global Definitions
 - Parameters 1
 - Materials
 - Component 1 (comp 1)
 - Definitions
 - Geometry 1**
 - Cylinder 1 (cy1)
 - Form Union (fin)
 - Materials
 - Laminar Flow (spf)
 - Fluid Properties 1
 - Initial Values 1
 - Wall 1
 - Transport of Diluted Species (tds)
 - Fluid 1
 - No Flux 1
 - Initial Values 1
 - Multiphysics
 - Mesh 1
 - Study 1
 - Step 1: Stationary
 - Results

Settings

Cylinder

Build Selected Build All Objects

Label: Cylinder 1

Object Type

Type: Solid

Size and Shape

Radius: $10\text{E-}6$ m

Height: $3\text{E-}5$ m

Position

x: 0 m

y: 0 m

z: 0 m

Axis

Axis type: z-axis

Rotation Angle

Rotation: 0 deg

Coordinate System

Work plane: xy-plane

Layers

Selections of Resulting Entities

☐ Resulting objects selection

Show in physics: Domain selection

Color: None

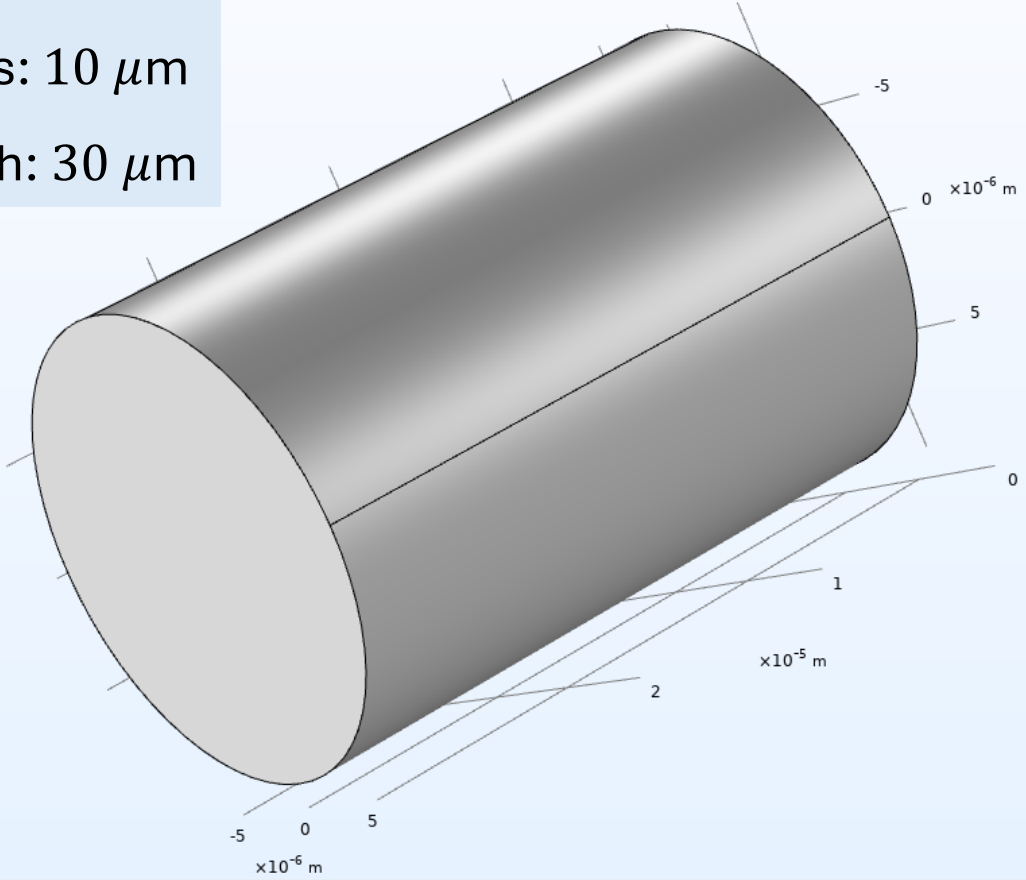
☐ Create layer selections

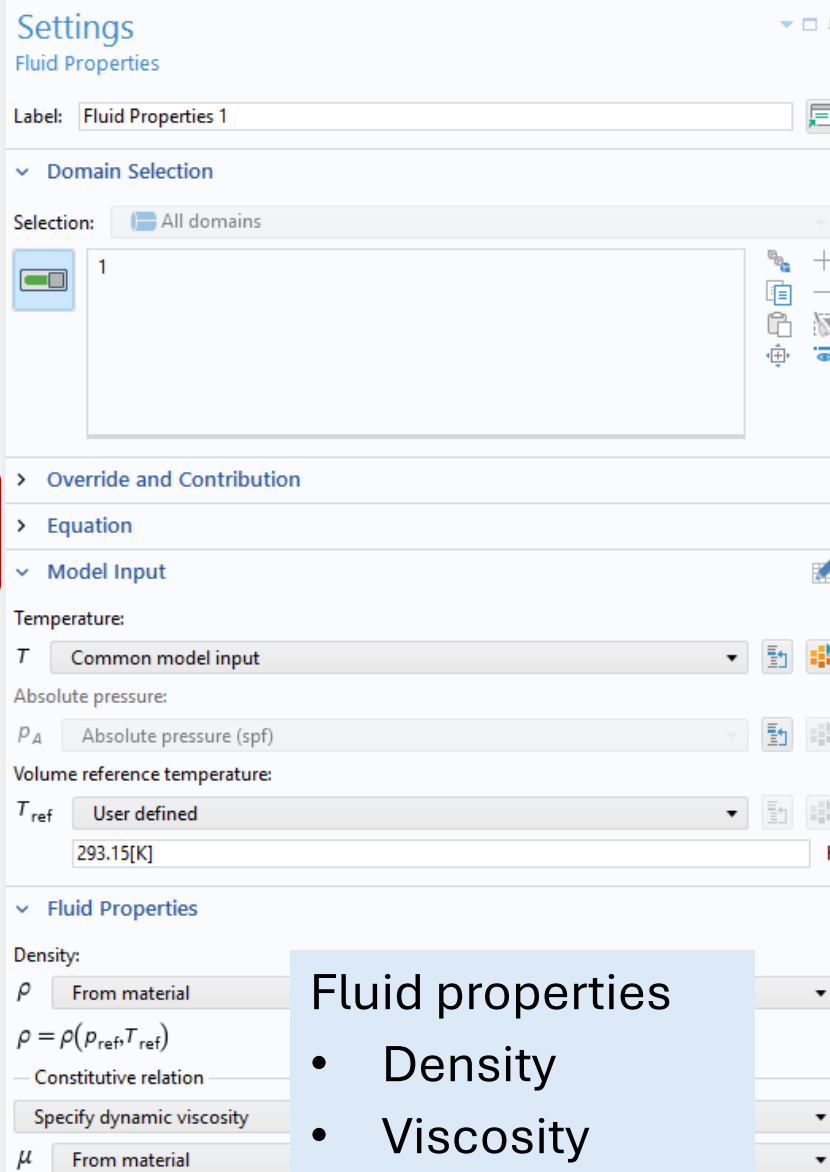
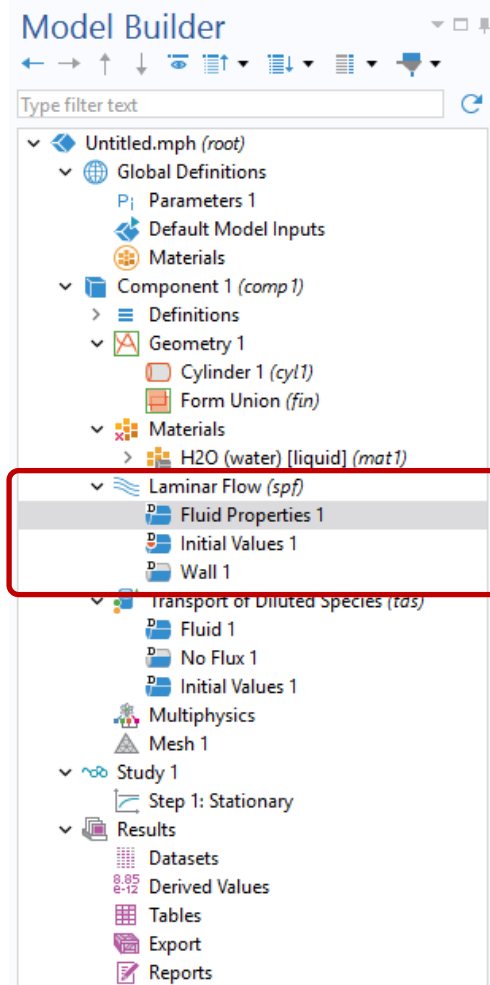
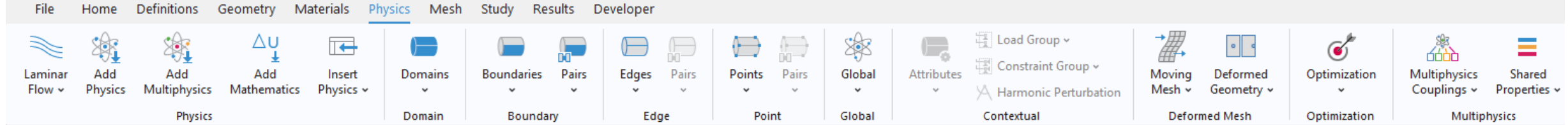
☒ Show in physics

Graphics

Cylinder:

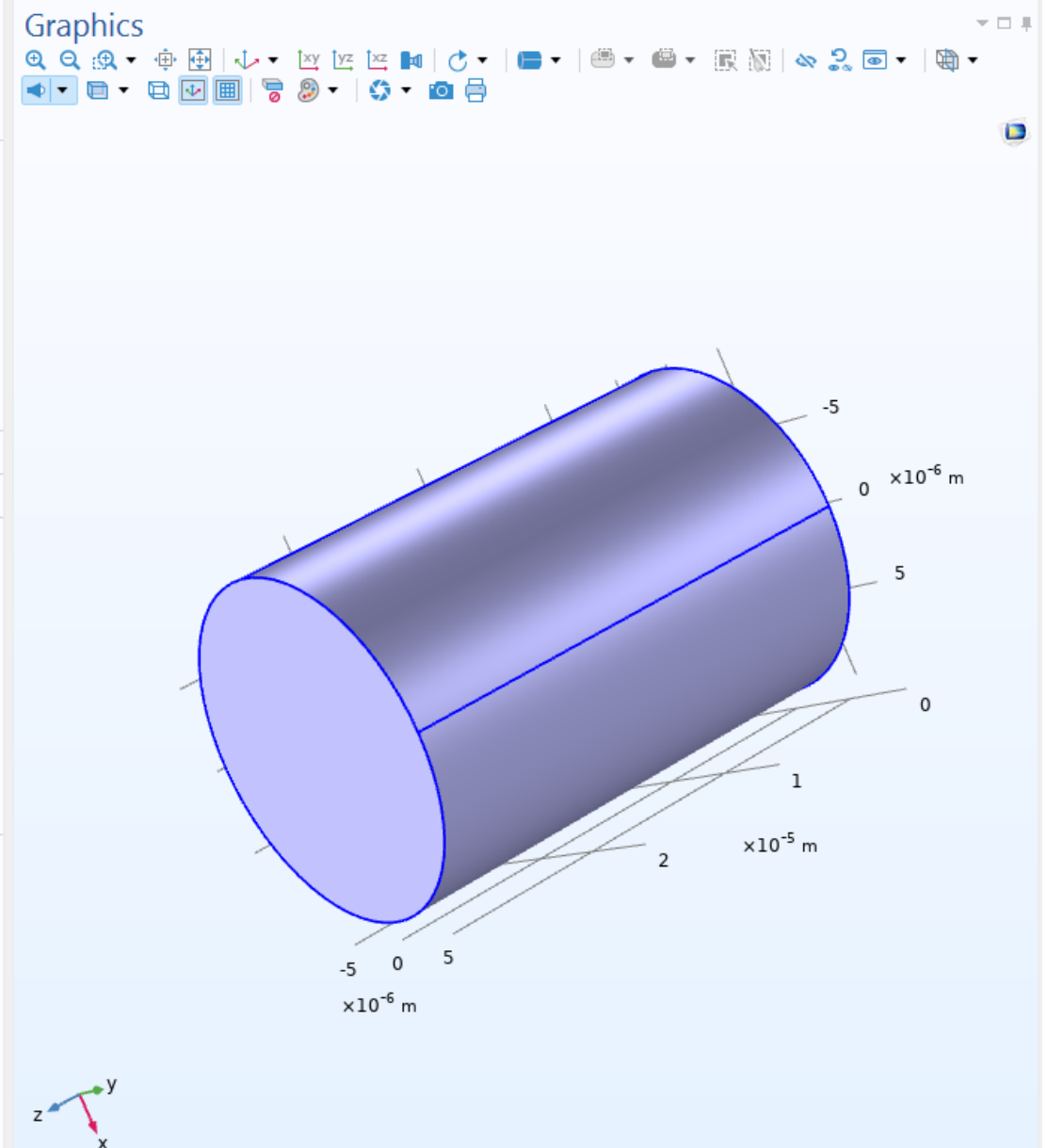
- Radius: $10\text{ }\mu\text{m}$
- Length: $30\text{ }\mu\text{m}$

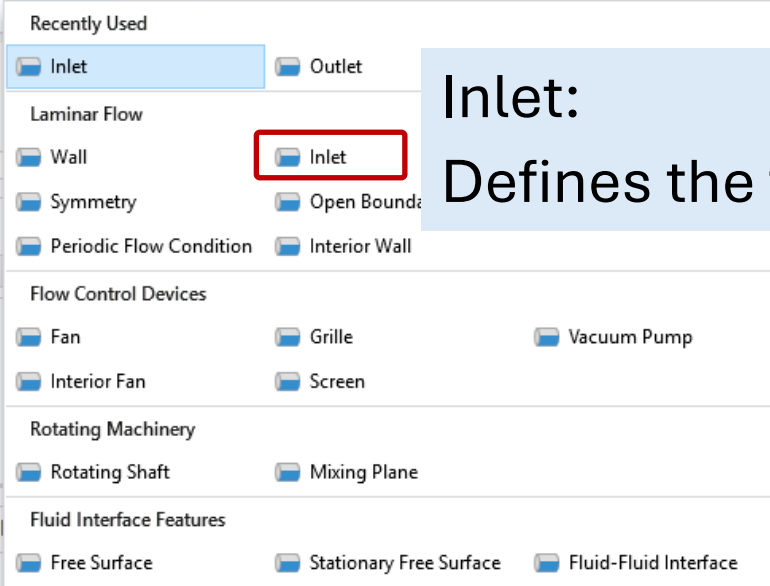
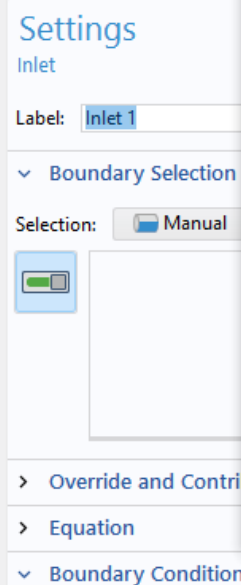
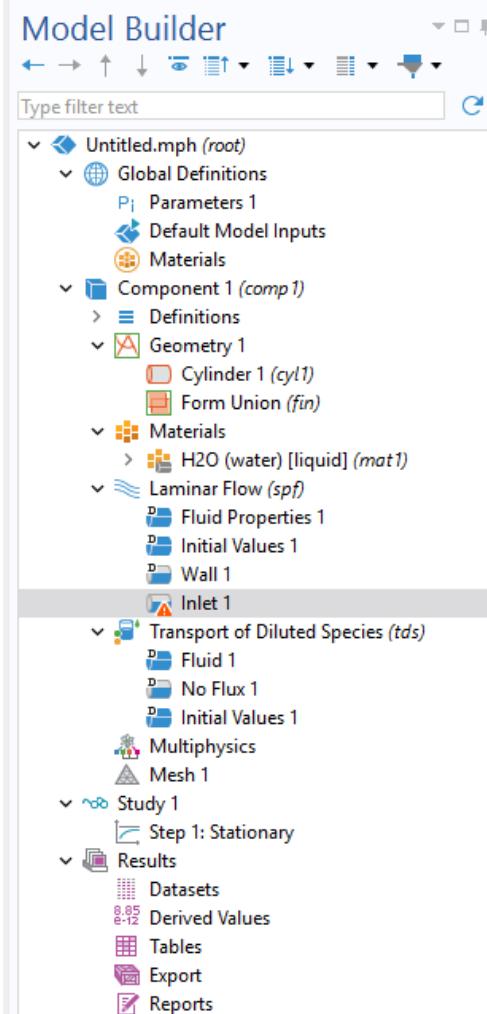
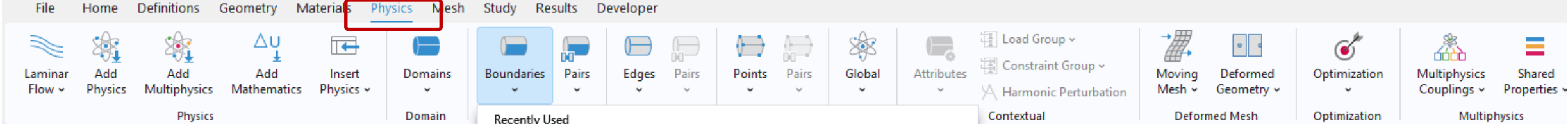




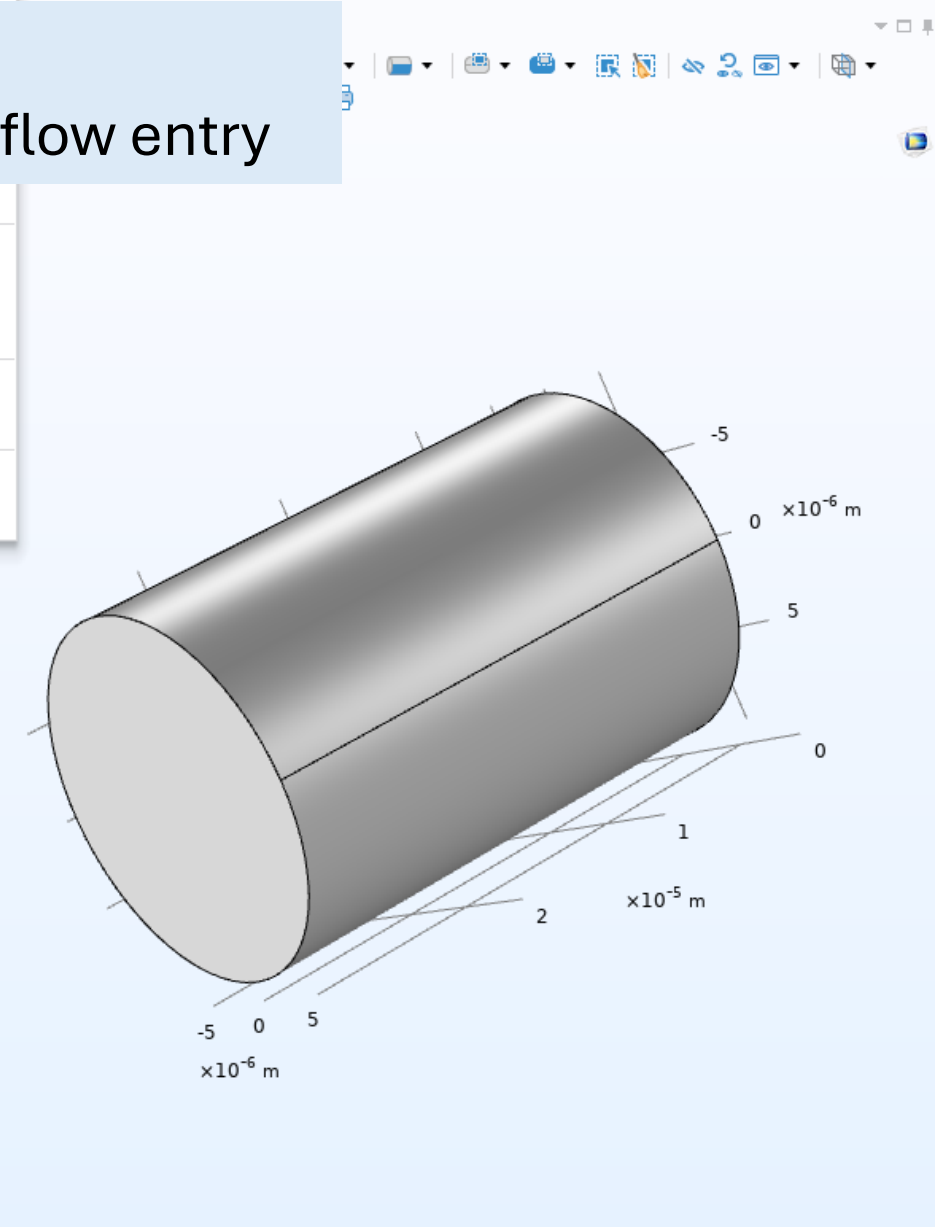
Fluid properties

- Density
- Viscosity





Inlet:
Defines the flow entry



Model Builder

- Type filter text
- ▼ Untitled.mph (root)
 - ▼ Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
 - ▼ Component 1 (comp1)
 - Definitions
 - ▼ Geometry 1
 - Cylinder 1 (cyl1)
 - Form Union (fin)
 - Materials
 - H2O (water) [liquid] (mat1)
 - ▼ Laminar Flow (spf)
 - Fluid Properties 1
 - Initial Values 1
 - Wall 1
 - Inlet 1**
 - Transport of Dilute Species (tas)
 - Fluid 1
 - No Flux 1
 - Initial Values 1
 - Multiphysics
 - Mesh 1
 - ▼ Study 1
 - Step 1: Stationary
 - ▼ Results
 - Datasets
 - 8.85 8.12 Derived Values
 - Tables
 - Export
 - Reports

Inlet

Settings

Inlet

Label: Inlet 1

▼ Boundary Selection

Selection: Manual

4

Inlet:
Side number 4

▼ Overview and Connections

▼ Equation

▼ Boundary Condition

Velocity

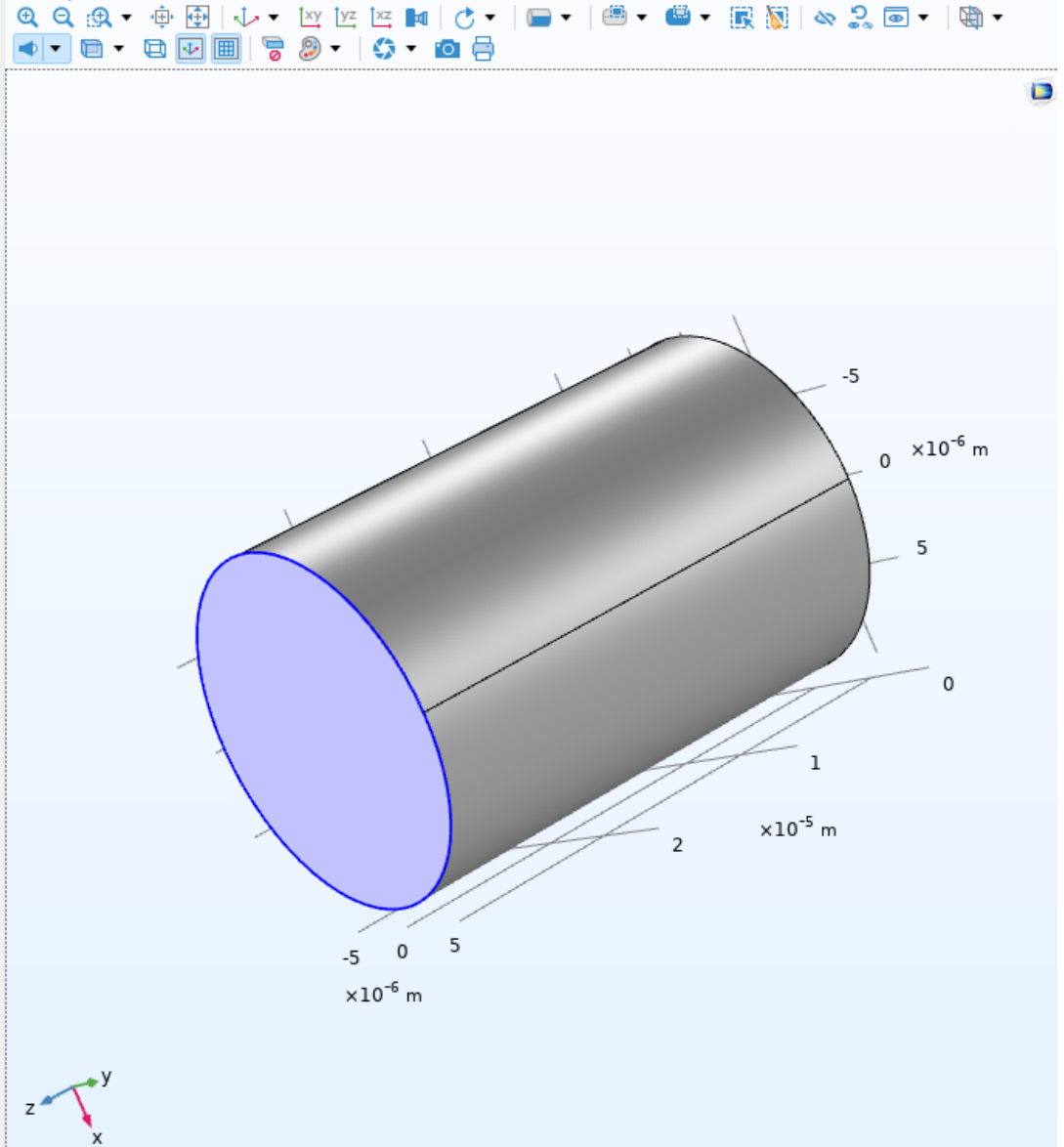
▼ Velocity

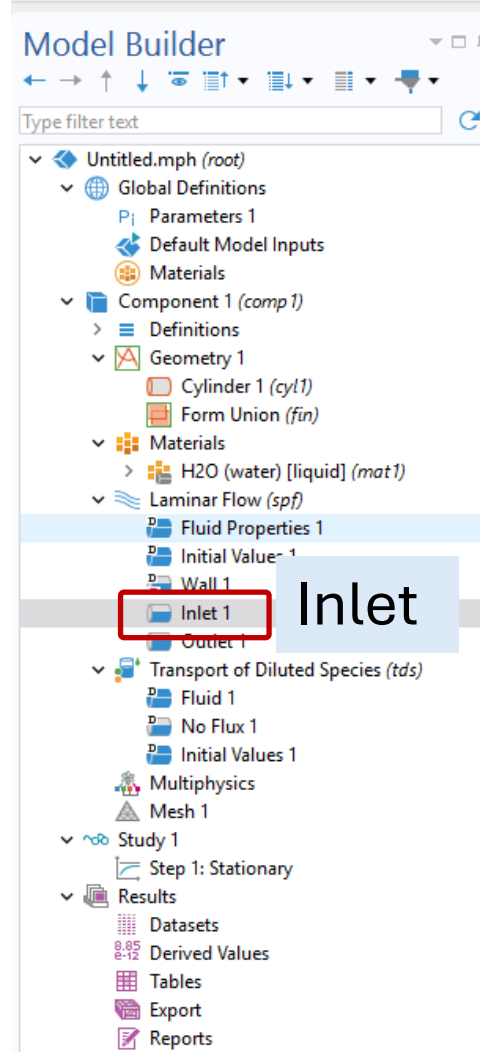
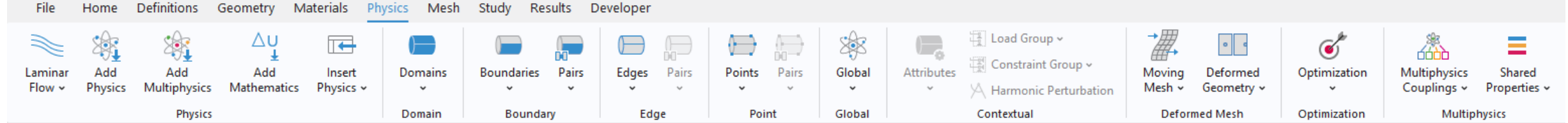
☒ Normal inflow velocity

☐ Velocity field

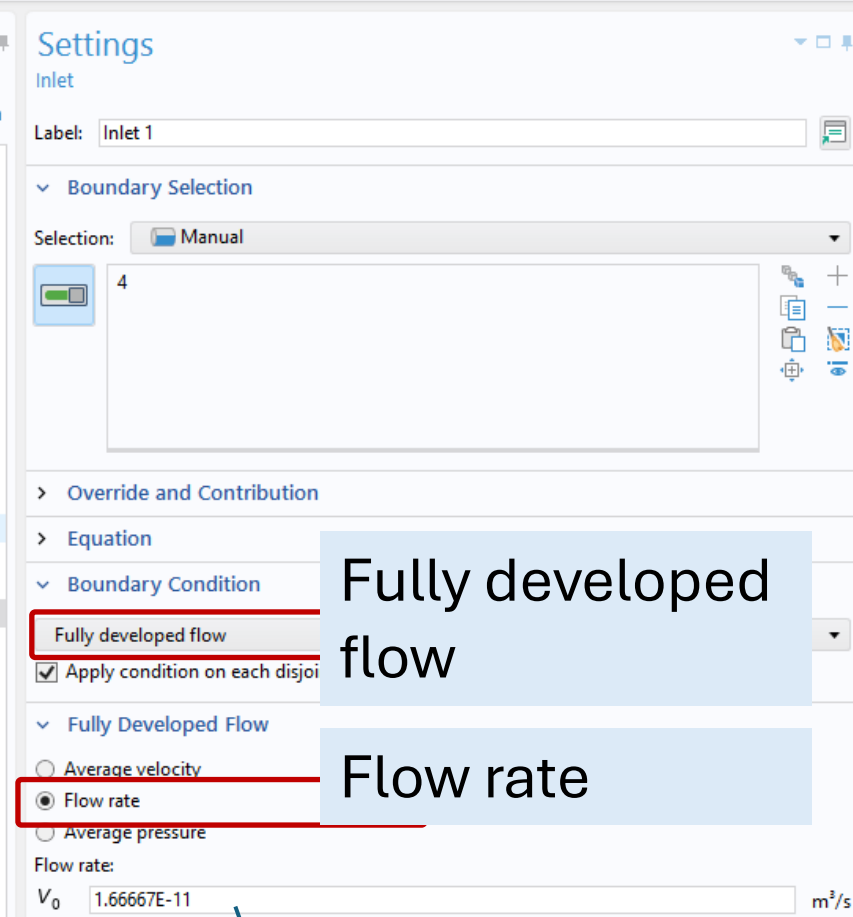
U_0 0 m/s

Graphics





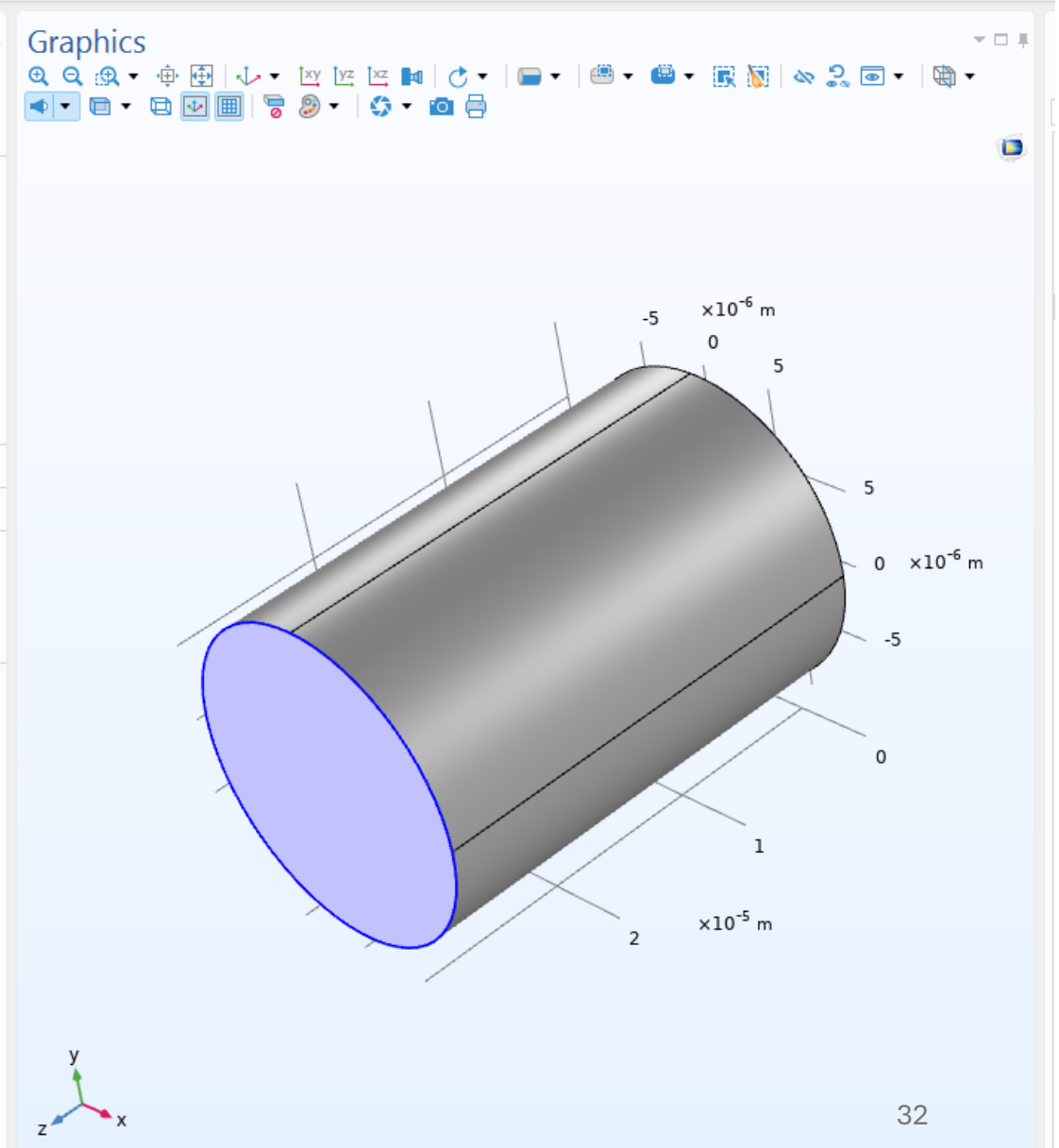
Inlet



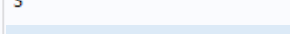
Fully developed flow

Flow rate

Flow rate:
 $Q = 1 \mu\text{L}/\text{min}$



Outlet



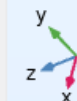
3

Outlet:

Side number 3

4. Boundary Condition

Pa



Model Builder

- Type filter text
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 - Initial Values 1
 - Wall 1
 - Inlet 1
 - Outlet 1
 - Transport of Diluted Species (tds)
 - Fluid 1
 - No
 - Init
 - Multipl
 - Mech 1
 - Study 1**
 - Step 1: Stationary**
 - Solver Configurations
 - Results
 - Datasets
 - Derived Values
 - Tables
 - Velocity (spf)**
 - Volume 1**
 - Pressure (spt)
 - Export
 - Reports

Settings

Stationary

Compute Update Solution

Label: Stationary

Study Settings

Tolerance: Physics controlled

Results While Solving

Physics and Variables Selection

☐ Modify model configuration for study step

Node	Solve for	Equation form
Component 1 (comp1)	☒	
Laminar Flow (spf)	☒	Automatic (Stationary)
Transport of Diluted Species (tds)	☐	Automatic (Stationary)

Values of Dependent Variables

Store in Output

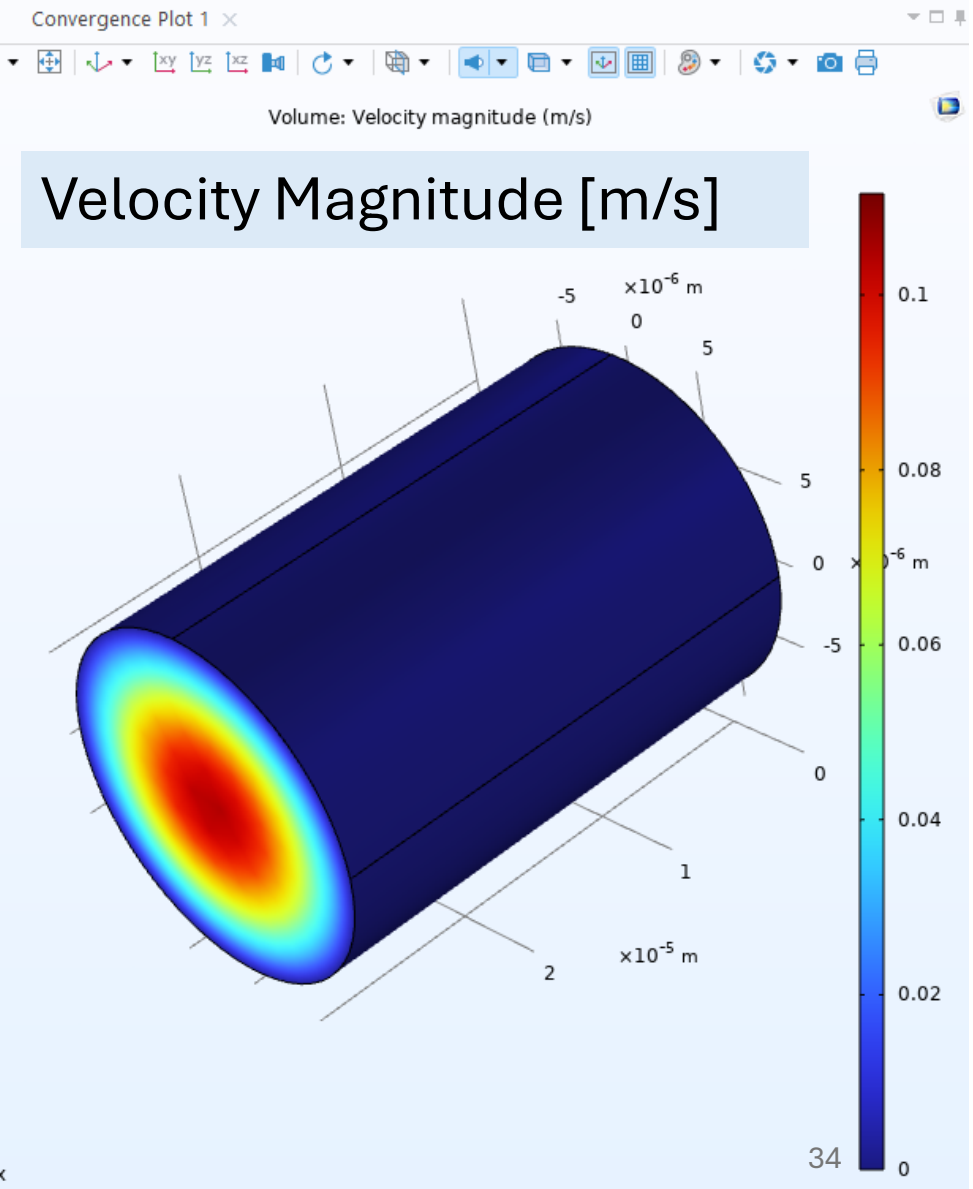
Mesh Selection

Stationary study

Solve for Laminar Flow only

Velocity Volume

Graphics



File Home Definitions Geometry Materials **Physics** Mesh Study Results Developer

Transport of Diluted Species Add Physics Add Multiphysics Add Mathematics Insert Physics Domains Boundaries Pairs Edges Pairs Points Pairs Global Attributes Load Group Constraint Group Moving Mesh Deformed Geometry Optimization Multiphysics Couplings

Model Builder

Settings

Fluid

Domain Selection

Selection: All domains

1

Override and Contribution

Equation

Model Input

Temperature:

T Common model input

Coordinate System Selection

Coordinate system: Global coordinate system

Convection

Velocity field:

U Velocity field (spf)

Diffusion

Source: Material

Material: NaOH [liquid] (mat2)

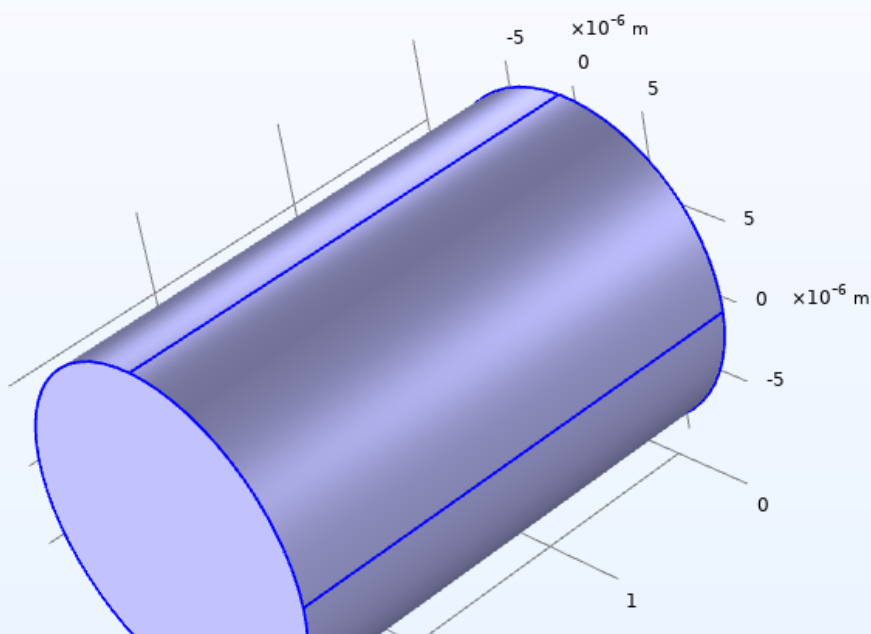
Diffusion coefficient:

D_c User defined

$2.125 \times 10^{-9} \text{ m}^2/\text{s}$

Graphics

Convergence Plot 1



3D visualization of a cylindrical domain with a mesh, showing the geometry and coordinate system.

- Velocity from Laminar Flow Physics
- Material: NaOH
- Diffusion coefficient: $2.125 \times 10^{-9} \text{ m}^2/\text{s}$

File Home Definitions Geometry Materials **Physics** Mesh Study Results Developer

Transport of Diluted Species Add Physics Add Multiphysics Add Mathematics Insert Physics Domains Boundaries Pairs Edges Pairs Points Pairs Global Attributes Load Group Constraint Group Harmonic Perturbation Moving Mesh Deformed Geometry Optimization Multiphysics Couplings

Model Builder Settings

Type filter text

- Untitled.mph (root)
 - Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
 - Component 1 (comp1)
 - Definitions
 - Rectangle 1 (rect1)
 - Analytic 1 (pulse)
 - Boundary System 1 (sys1)
 - View 1
 - Geometry 1
 - Cylinder 1 (cyl1)
 - Form Union (fin)
 - Materials
 - H2O (water) [liquid] (mat1)
 - NaOH [liquid] (mat2)
 - Basic (def)
 - Thermal expansion (ThermalE)
 - Piecewise (dL)
 - Piecewise 1 (CTE)
 - Laminar Flow (spf)
 - Fluid Properties 1
 - Initial Values 1
 - Wall 1
 - Inlet 1
 - Outlet 1
 - Transport of Diluted Species (tds)
 - Fluid 1
 - No Flux 1
 - Initial values 1
 - Inflow 1**
 - Outflow 1
 - Multiphysics
 - Mesh 1
 - Study 1
 - Step 1: Stationary
 - Solver Configurations
 - Results

Settings

Inflow

Label: Inflow 1

Boundary Selection

Selection: Manual

4

Recently Used

- Outflow
- Inflow
- Transport of Diluted Species
 - Inflow
 - Outflow
 - Concentration
 - Symmetry
 - Periodic Condition
- Flux
- No Flux
- Open Boundary
- Partition Condition
- Flux Discontinuity
- Surface Reactions
 - Surface Reactions
 - Fast Irreversible Surface Reaction
 - Electrode Surface Coupling
- Interior Surfaces
 - Thin Diffusion Barrier
 - Thin Impermeable Barrier

Override and Contribution

Equation

Concentration

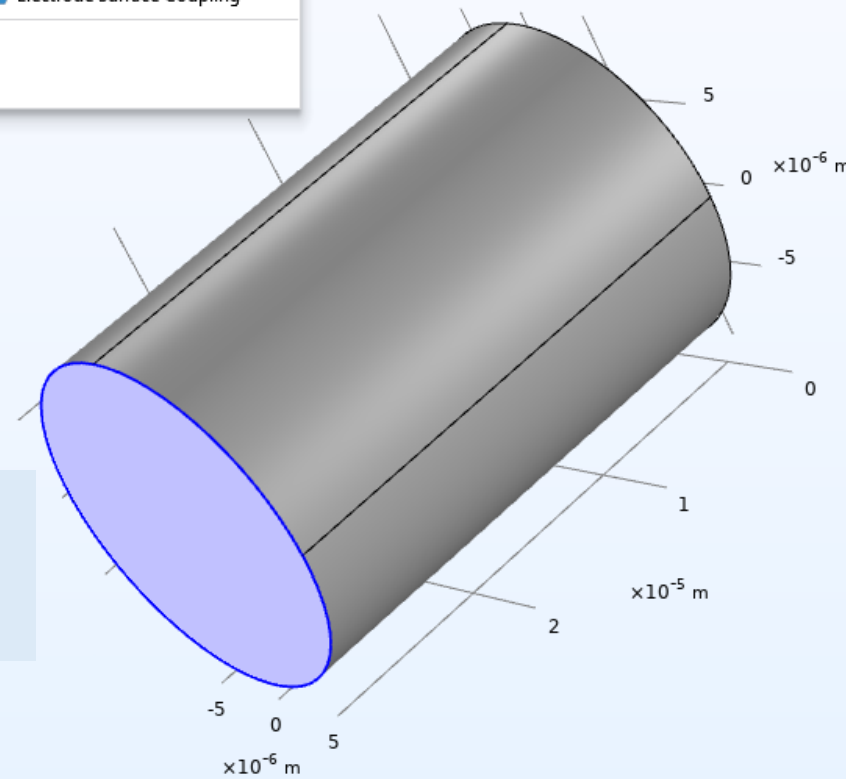
$c_{0,c}$ 1 mol/m³

Boundary Condition Type

Concentration constraint

Constant concentration with time

$c = 1 \text{ mol/m}^3$



Model Builder

Type filter text

- Untitled.mph (root)
 - Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
 - Component 1 (comp 1)**

- Functions
 - Analytic
 - Interpolation
 - Piecewise
 - Gaussian Pulse
 - Ramp
 - Rectangle**
 - Step
 - Triangle
 - Waveform
 - Normal Distribution
 - Random
 - Elevation (DEM)
 - Image
 - Least-Squares Fit
 - Gaussian Process
 - Polynomial Chaos Expansion
 - Deep Neural Network
 - Partial Fraction Fit
 - Function Switch

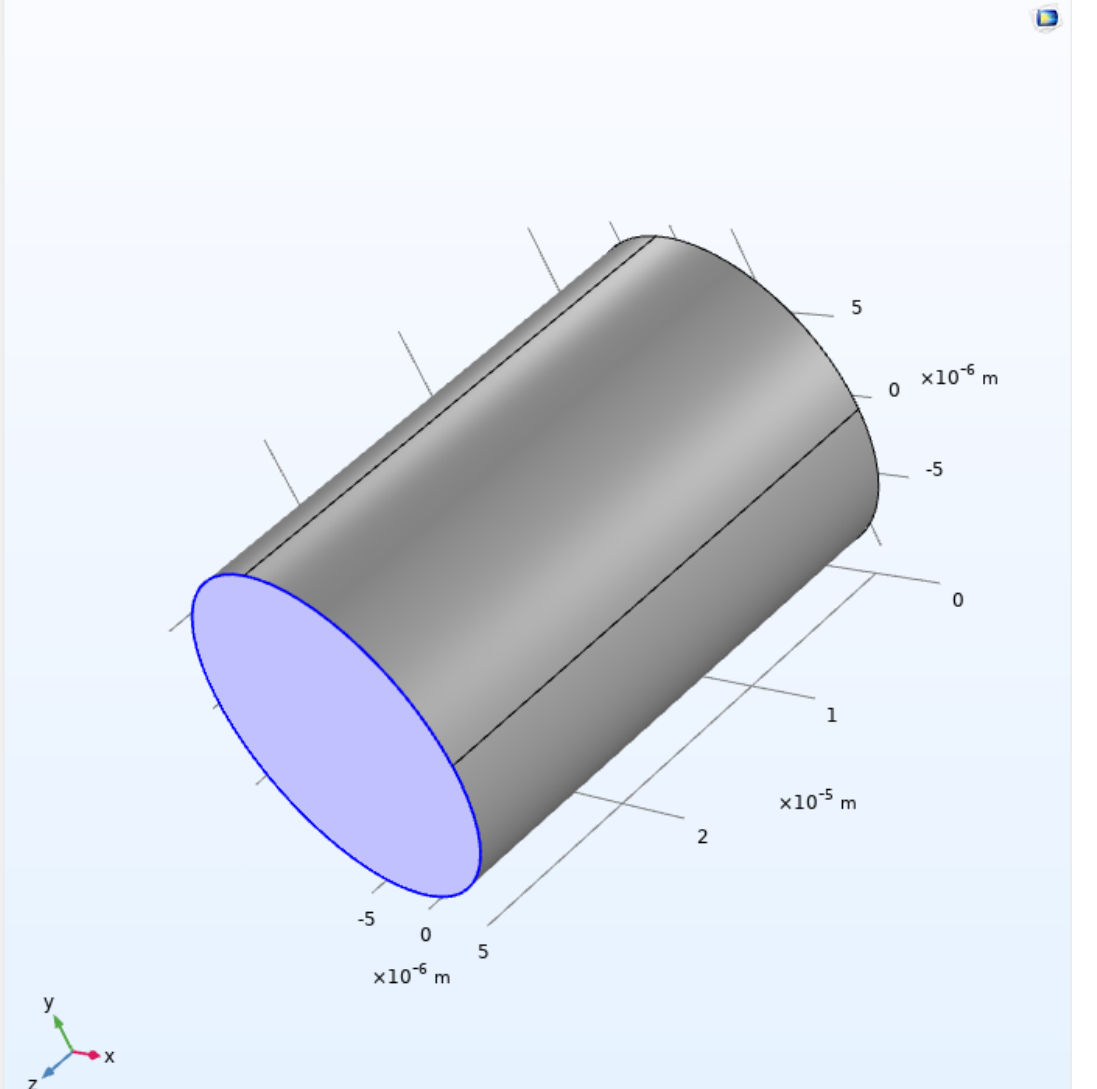
Settings

Definitions

Add a Rectangle function

Graphics

Convergence Plot 1



Type filter text

 Plot Create Plot

Label: Rectangle 1

Function name: rect1

Parameters

Lower limit: 0

Upper limit: 1

Baseline: 0

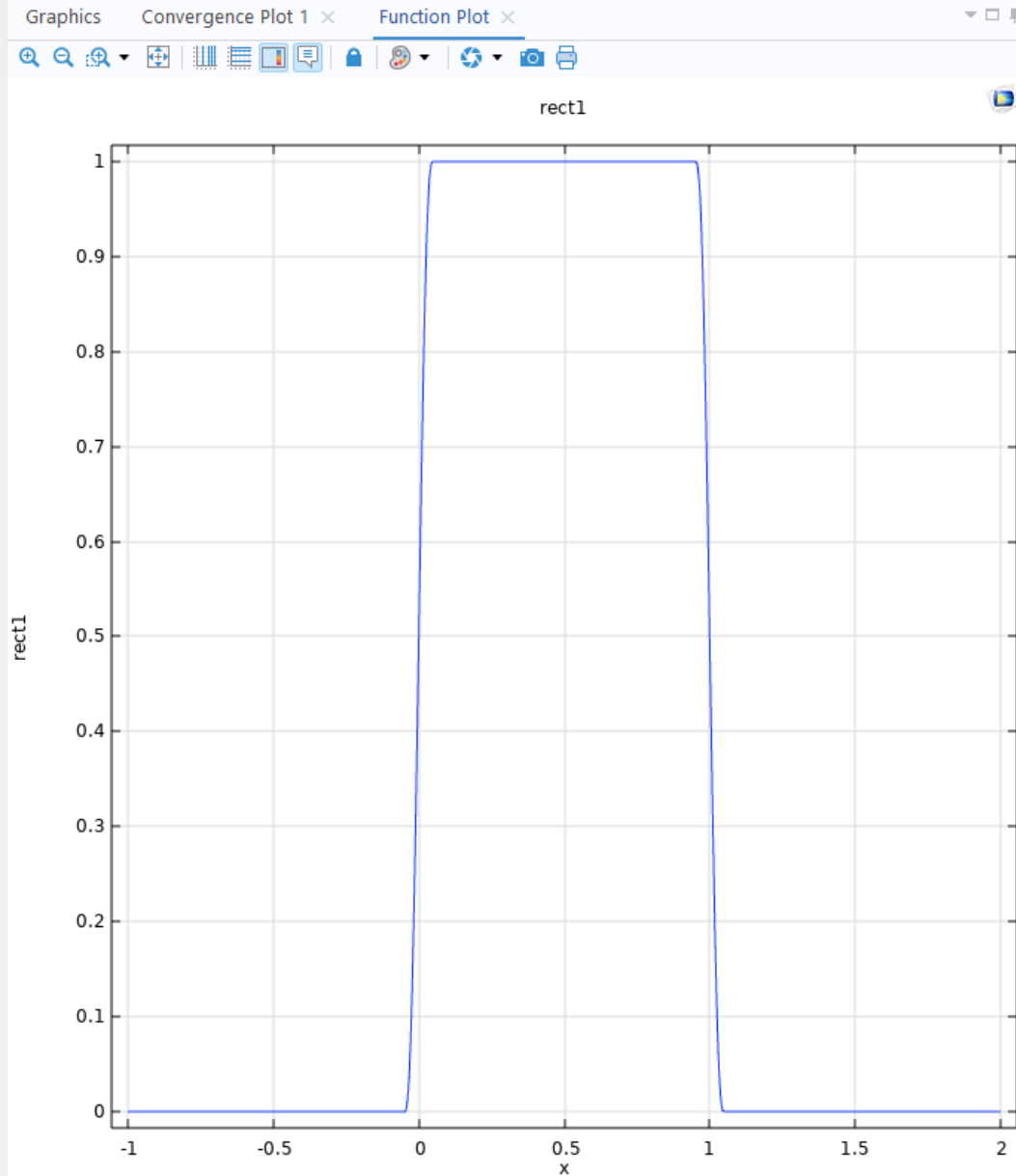
Amplitude: 1

- Smoothing

Size of transition zone: ☒ 0.1

Number of continuous derivatives: 2

Duration: 1



Model Builder

Type filter text

- Untitled.mph (root)
 - Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
 - Component 1 (comp 1)
 - Definitions
 - Rect
 - Ana
 - Bou
 - View
 - Geomet
 - Cylind
 - Form
 - Material
 - H2O
 - NaC
 - Laminar
 - Fluid
 - Initia
 - Wall
 - Inlet
 - Outl
 - Transpo
 - Fluid
 - No f
 - Initia
 - Inflc
 - Outl

Multiphysics Mesh 1

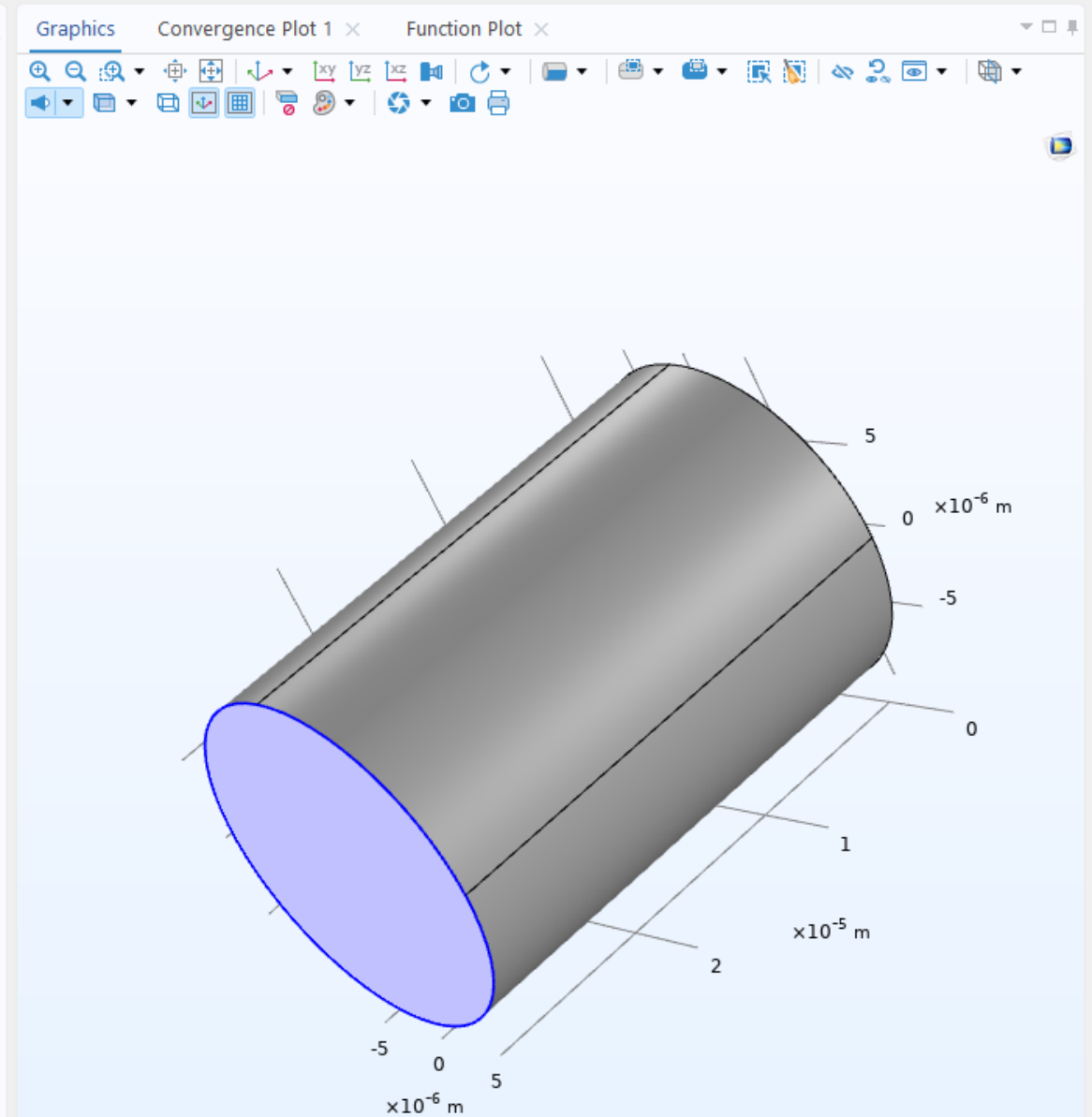
Settings

Definitions

Add an Analytic function

Functions

- Interpolation
- Piecewise
- Gaussian Pulse
- Ramp
- Rectangle
- Step
- Triangle
- Waveform
- Normal Distribution
- Random
- Elevation (DEM)
- Image
- Least-Squares Fit
- Gaussian Process
- Polynomial Chaos Expansion
- Deep Neural Network
- Partial Fraction Fit
- Function Switch



Model Builder

Type filter text

- Untitled.mph (root)
 - Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
 - Component 1 (comp 1)
 - Definitions
 - Rectangle 1 (rect1)
 - Analytic 1 (pulse)
 - Boundary System 1 (sys1)
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 - Cylinder 1 (cyl1)
 - Form Union (fin)
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 - NaOH [liquid] (mat2)
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 - Thermal expansion (ThermalE)
 - Piecewise (dL)
 - Piecewise 1 (CTE)
 - Laminar Flow (spf)
 - Fluid Properties 1
 - Initial Values 1
 - Wall 1
 - Inlet 1
 - Outlet 1
 - Transport of Diluted Species (tds)
 - Fluid 1
 - No Flux 1
 - Initial Values 1
 - Inflow 1
 - Outflow 1
 - Multiphysics
 - Mesh 1
 - Study 1
 - Step 1: Stationary
 - Solver Configurations
 - Results
 - Datasets

Settings

Analytic

Plot Create Plot

Label: Analytic 1

Function name: pulse

Definition

Expression: rect1(t)

Arguments: t

Derivatives: Automatic

Periodic Extension

Units

Function:

Argument	Unit
t	s

Local Parameters

Advanced

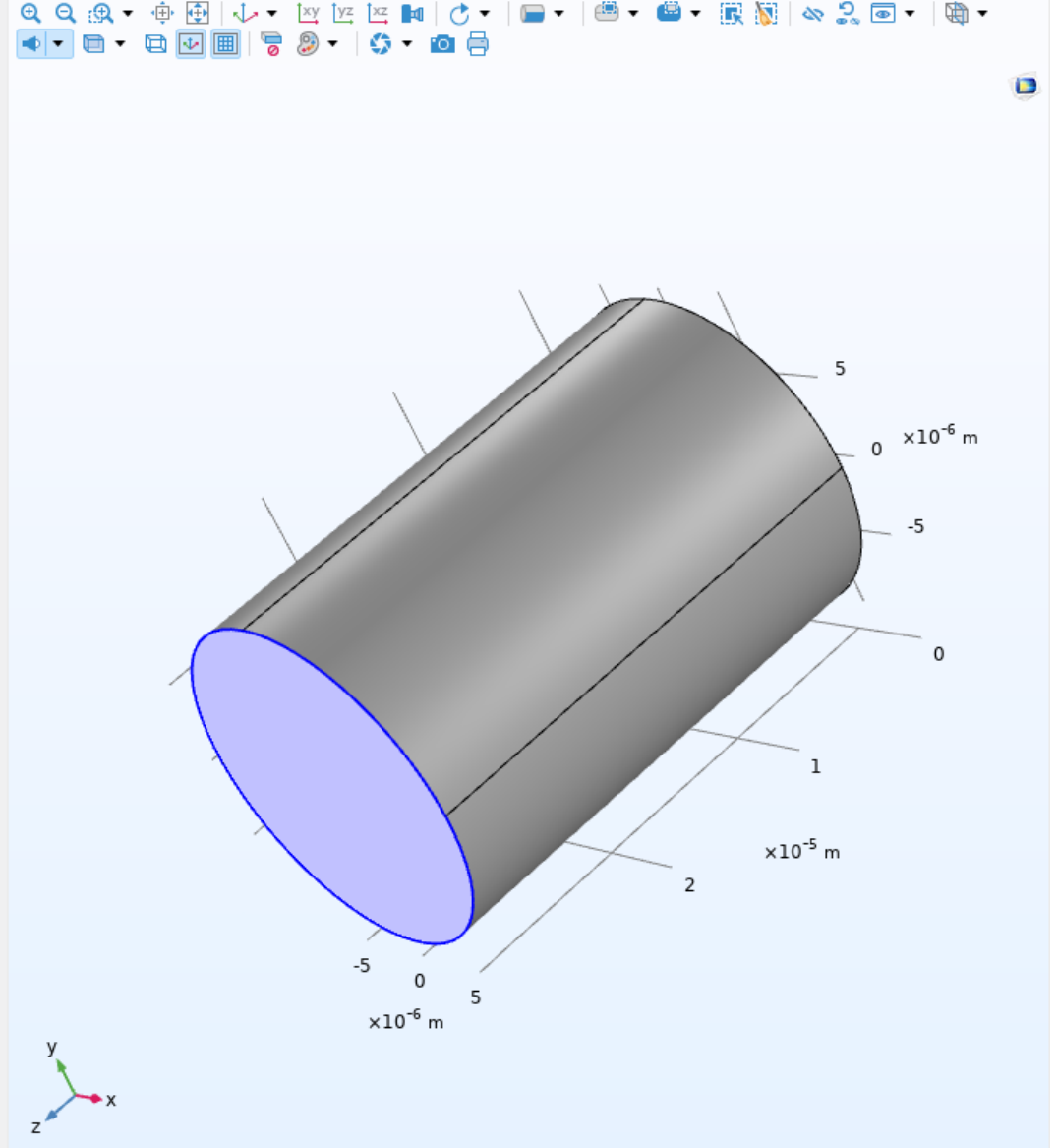
Plot Parameters

Plot	Argument	Lower limit	Upper limit	Fixed value	Unit
☒	t	0	1	0	s

Analytic function

- Named as „pulse“
- Calls to rect1(t)

Graphics



Model Builder

Type filter text

- Untitled.mph (root)
 - Global Definitions
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 - Component 1 (comp 1)
 - Definitions
 - Rectangle 1 (rect 1)
 - Analytic 1 (pulse)
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 - No Flux 1
 - Initial Values 1
 - Inflow 1
 - Outflow 1
 - Multiphysics
 - Mesh 1
 - Study 1
 - Step 1: Stationary
 - Solver Configurations
 - Results
 - Datasets

Settings

Inflow

Label: Inflow 1

Boundary Selection

Selection: Manual

4

Override and Contribution

Equation

Concentration

$c_{0,c}$ $10 \cdot \text{pulse}(t)$ mol/m³

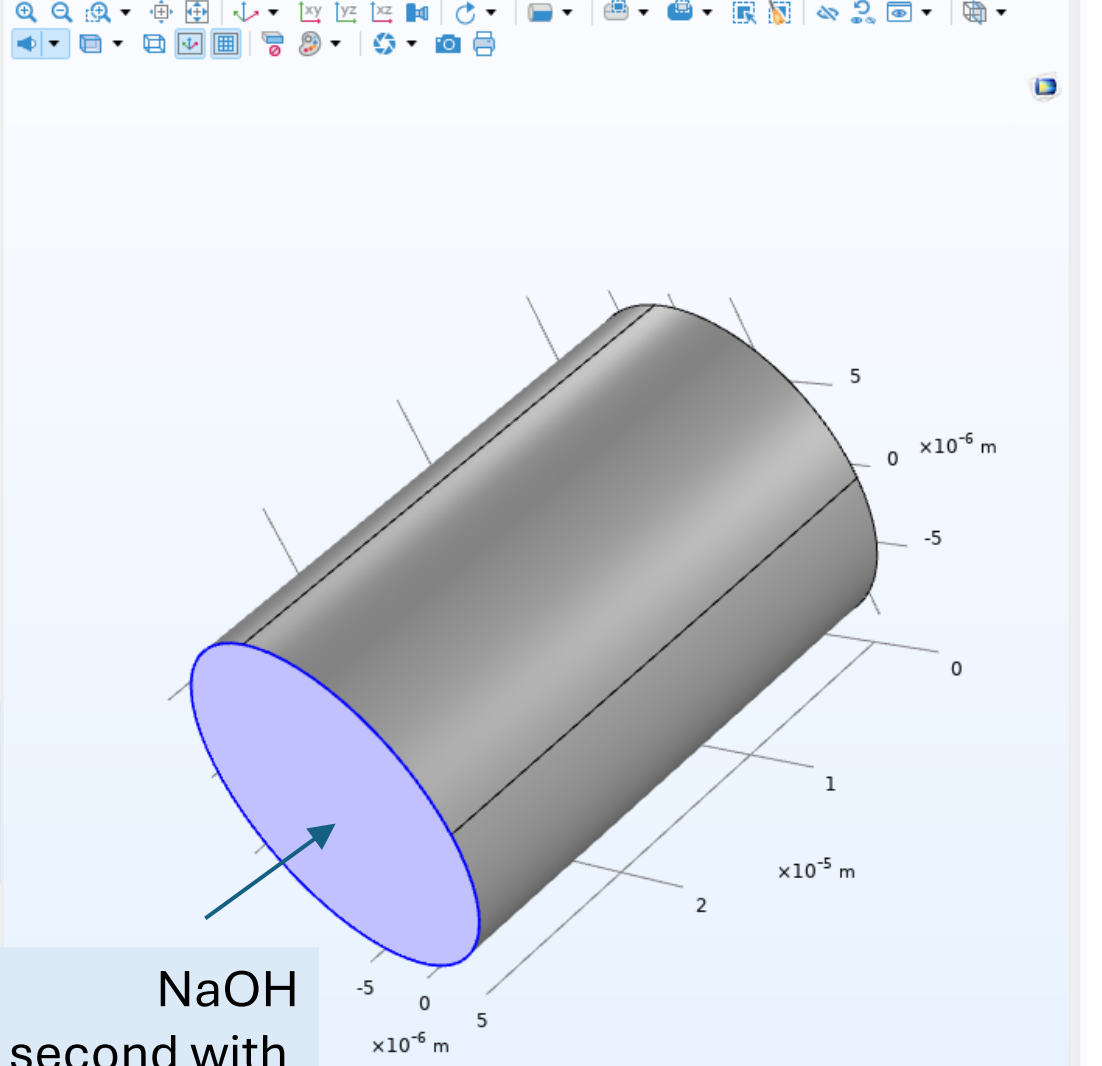
Boundary Condition Type

Concentration constraint

Call to the „pulse“ function

- $10 \cdot \text{pulse}(t)$

Graphics



NaOH appears for 1 second with concentration 10 mol/m³

File Home Definitions Geometry Material **Physics** Mesh Study Results Developer

Laminar Flow Add Physics **Add Multiphysics** Add Mathematics Insert Physics Domains Boundaries Pairs Edges Pairs Points Pairs Global Attributes Load Group Constraint Group Moving Mesh Deformed Geometry Optimization Multiphysics Couplings Shared Properties Thermodynamics

Model Builder

- single_pipe.mph (root)
 - Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
 - Component 1 (comp 1)
 - Definitions
 - Geometry 1
 - Materials
 - Laminar Flow (spf)
 - Transport of Diluted Species (tds)
 - Multiphysics**
 - Reacting Flow, Diluted Species 1 (rfd)**
 - Mesh 1
 - Study 1
 - Study 2
 - Step 1: Time Dependent
 - Solver Configurations
 - Results
 - Datasets
 - Derived Values
 - Tables
 - Velocity (spf)
 - Pressure (spf)
 - Concentration, Streamline (tds)
 - Concentration, Surface (tds)
 - 1D Plot Group 5
 - Table Graph 1
 - Export
 - Reports

Settings

Reacting Flow, Diluted Species

Label: Reacting Flow, Diluted Species 1

Name: rfd1

Domain Selection

Selection: All domains

1

Coupled Interfaces

Fluid flow: Laminar Flow (spf)

Species transport: Transport of Diluted Species (tds)

Graphics

0 5 $\times 10^{-6}$ m

$\times 10^{-5}$ m

1 2

5 0 $\times 10^{-6}$ m

-5

x y z

Select "Laminar Flow, Diluted Species"

Add Study Add Multip... Add Material

Add to Component

- Chemical Species Transport
 - Reacting Flow
 - Laminar Flow, Diluted Species**

Select the physics interfaces you want to couple

Physics	Couple
Laminar Flow (spf)	☒
Transport of Diluted Species (tds)	☒

Multiphysics couplings in study

Studies	Solve
Study 1	☒
Study 2	☒

Messages Progress Log Table 1

File Home Definitions Geometry Materials Physics Mesh **Study** Results Developer

Compute Study 2 **Add Study** Continue Update Solution Get Initial Value

Show Default Solver Show Default Plots Reset Default Plots Solver

Study Steps Parametric Sweep Function Sweep Material Sweep Combine Solutions Study Reference Optimization Uncertainty Quantification Surrogate Model Training Operations Evaluate Clear Solutions Clear All Solutions Cosimulation for Simulink External Interface

Model Builder

Type filter text

- single_pipe.mph (root)
 - Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
 - Component 1 (comp1)
 - Definitions
 - Geometry 1
 - Materials
 - Laminar Flow (spf)
 - Transport of Diluted Species (tds)
 - Multiphysics
 - Reacting Flow, Diluted Species 1 (rfd)
 - Mesh 1
 - Study 1
 - Study 2**
 - Step 1: Time Dependent**
 - Solver Configurations
 - Results
 - Datasets
 - Derived Values
 - Tables
 - Velocity (spf)
 - Pressure (spf)
 - Concentration, Streamline (tds)
 - Concentration, Surface (tds)
 - 1D Plot Group 5
 - Table Graph 1
 - Export
 - Reports

Settings

Time Dependent

Compute Update Solution

Label: Time Dependent

Time unit: s

Output times: range(0,0.1,5) s

Tolerance: Physics controlled

Results While Solving

Physics and Variables Selection

☐ Modify model configuration for study step

Node	Solve for	Equation form
Component 1 (comp1)	☒	
Laminar Flow (spf)	☐	Automatic (Stationary)
Transport of Diluted Species (tds)	☒	Automatic (Time dependent)
Multiphysics		
Reacting Flow, Diluted Species 1 (...)	☒	Automatic (Time dependent)

Values of Dependent Variables

Initial values of variables solved for

Settings: User controlled

Method: Initial expression

Study: Study 1, Stationary

Solution: Current (Solution 1 (sol1))

Selection: Automatic (single solution)

Values of variables not solved for

Settings: User controlled

Method: Initial expression

Study: Study 1, Stationary

Solution: Current (Solution 1 (sol1))

Selection: Automatic (single solution)

Graphics

Time=0 s

Molar concentration, c (mol/m³)

Time: 0s

Select "Time Dependent"

Study

Add Study Add Multip... Add Material

+ Add Study

Studies

- General Studies
 - Stationary
 - Time Dependent**
 - Empty Study

Physics interfaces in study

Interface	Solve
Physics	☐
Laminar Flow (spf)	☐
Transport of Diluted Species (tds)	☒

Multiphysics couplings in study

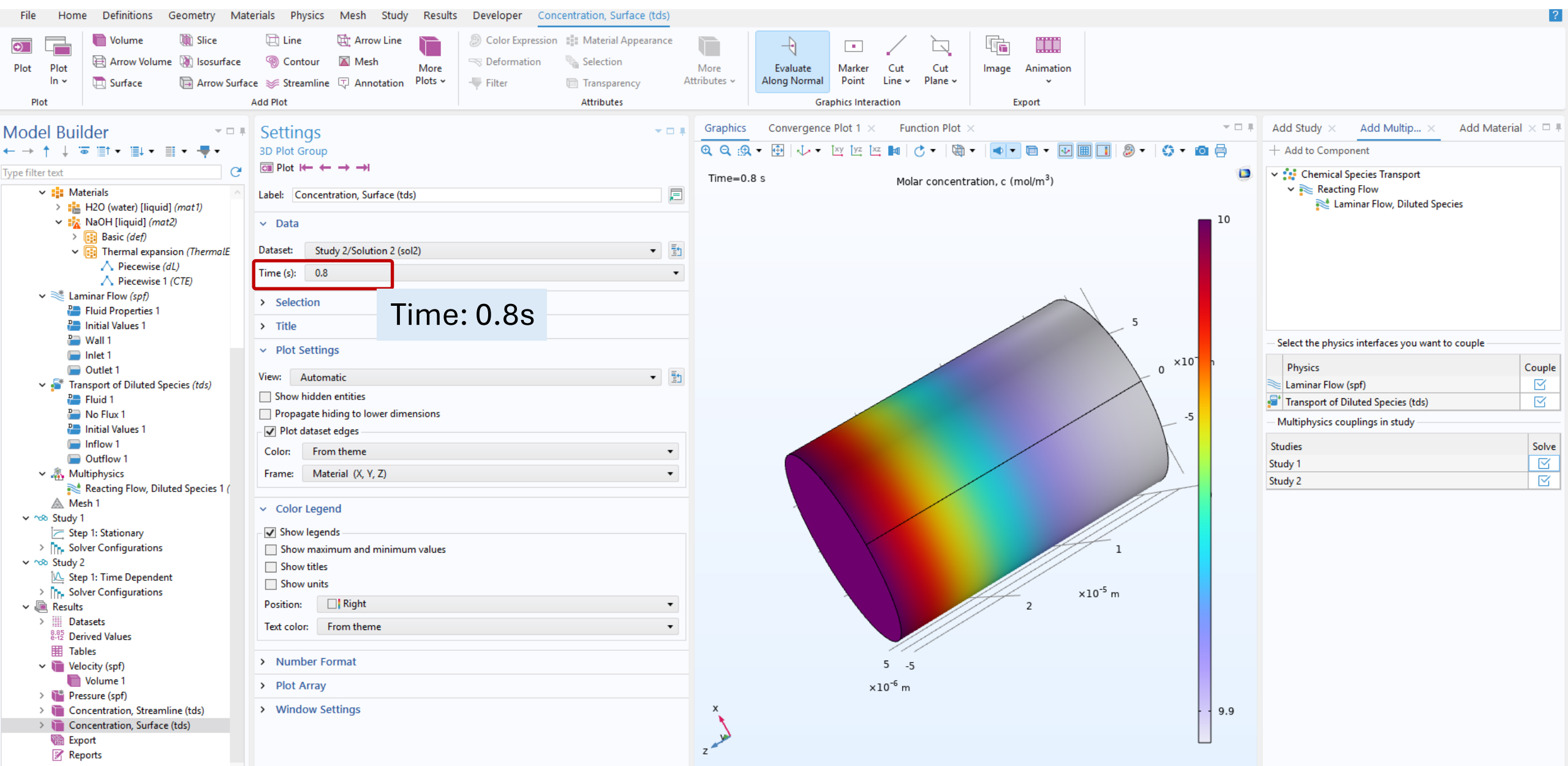
Coupling	Solve
Multiphysics couplings	☐
Reacting Flow, Diluted Species 1 (rfd1)	☒

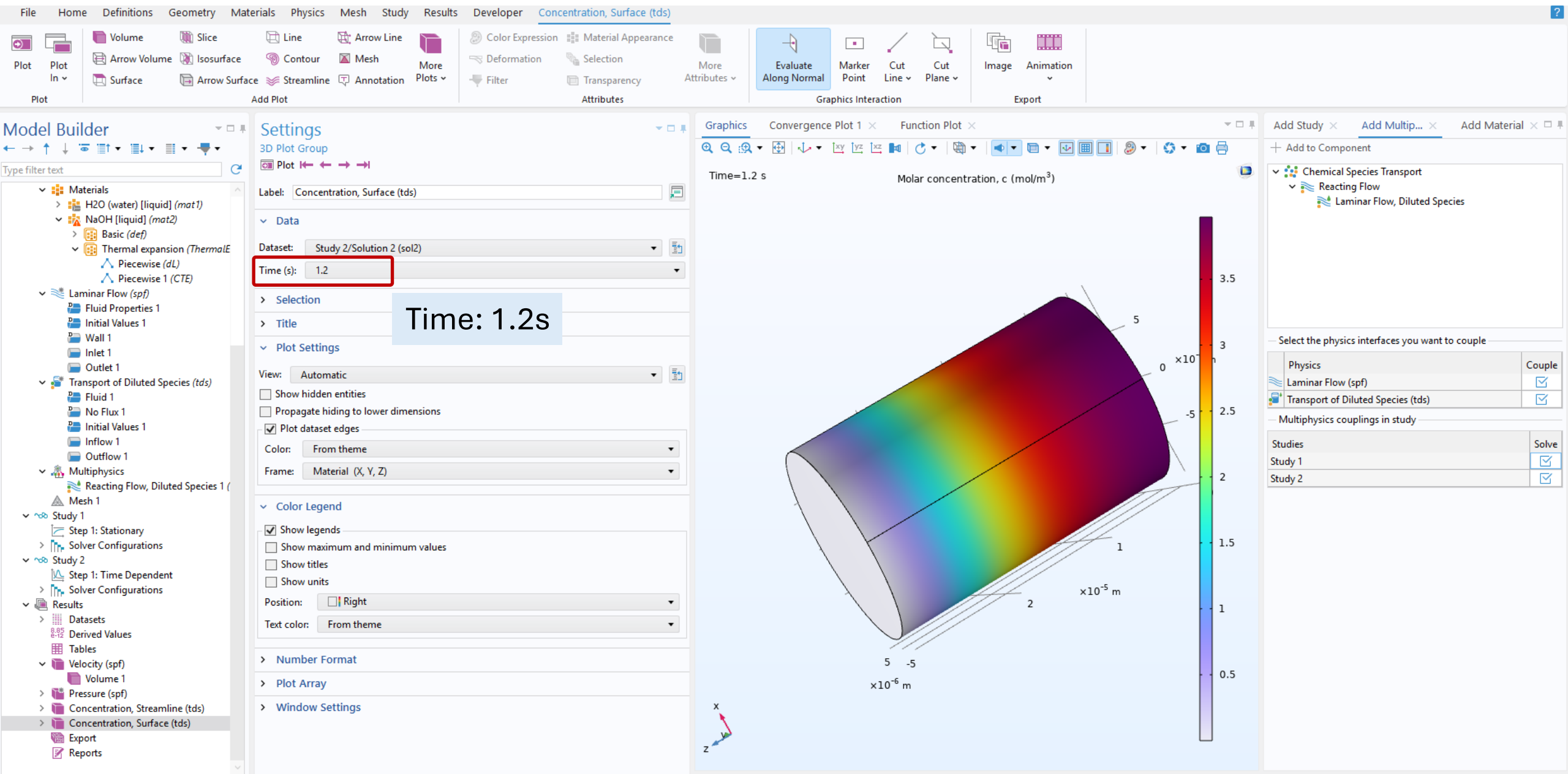
Messages

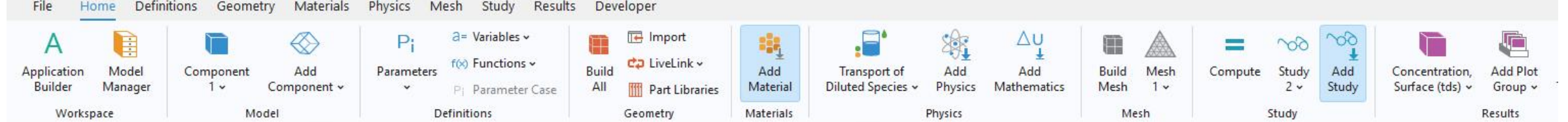
Progress Log Table 1

COMSOL Multiphysics 6.3.0.290

[Sep 15, 2026, 3:53 PM] Opened file: D:\papers\tutorial2026nanocom\single_pipe.mph

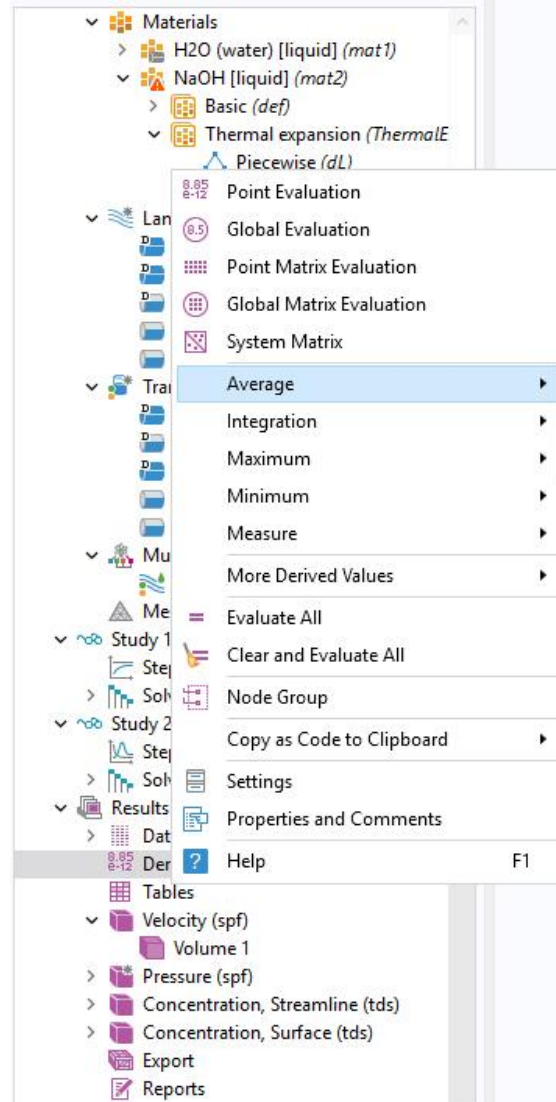






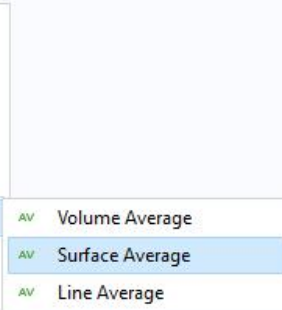
Model Builder

Type filter text



Settings

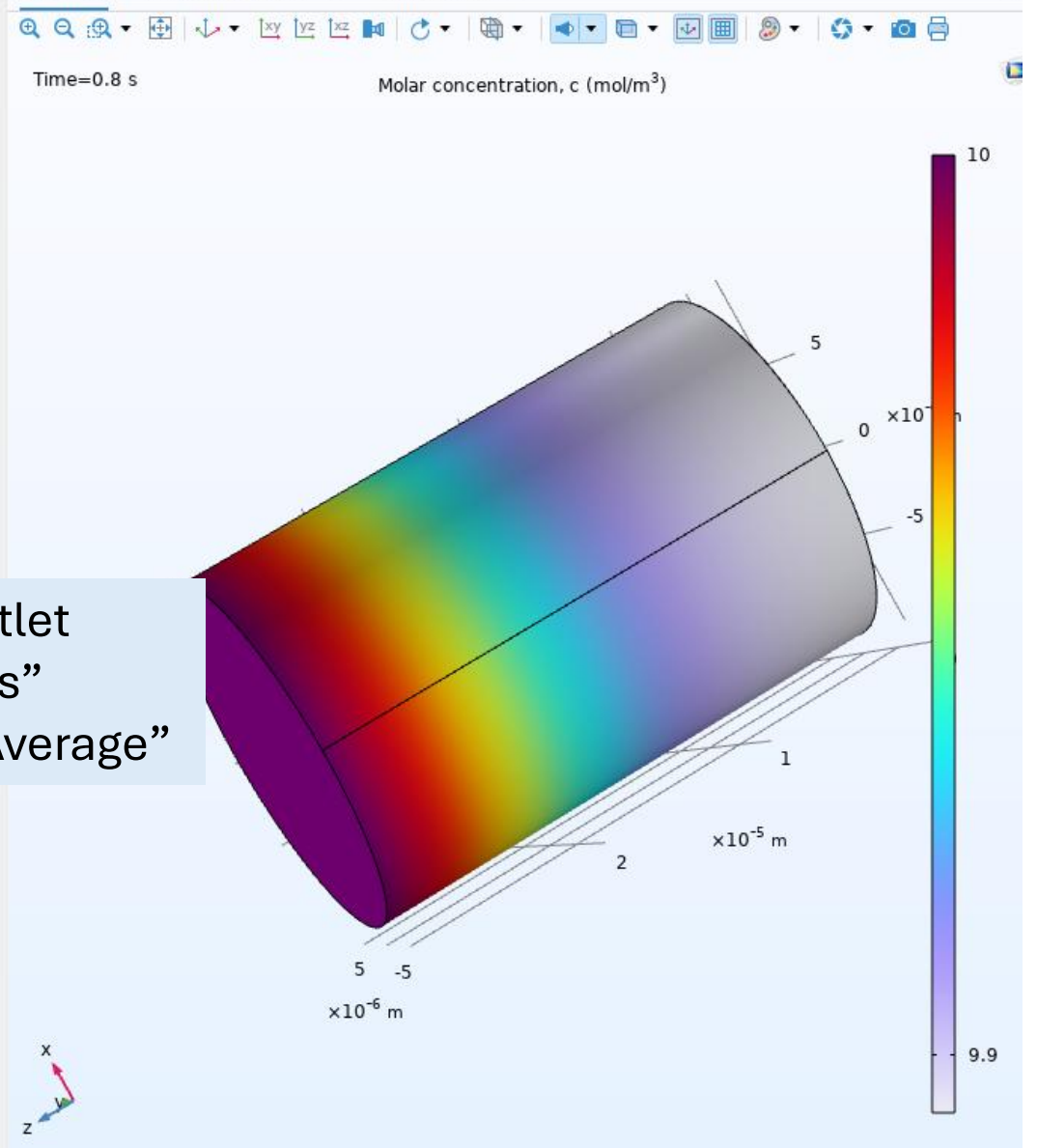
Derived Values



Average concentration at the Outlet

- Right-click on “Derived Values”
- Select “Average” → “Surface Average”

Graphics



Model Builder

Type filter text

- H2O (water) [liquid] (mat1)
- NaOH [liquid] (mat2)
 - Basic (def)
 - Thermal expansion (ThermalE)
 - Piecewise (dL)
 - Piecewise 1 (CTE)
- Laminar Flow (spf)
 - Fluid Properties 1
 - Initial Values 1
 - Wall 1
 - Inlet 1
 - Outlet 1
- Transport of Diluted Species (tds)
 - Fluid 1
 - No Flux 1
 - Initial Values 1
 - Inflow 1
 - Outflow 1
- Multiphysics
 - Reacting Flow, Diluted Species 1
- Mesh 1
- Study 1
 - Step 1: Stationary
 - Solver Configurations
- Study 2
 - Step 1: Time Dependent
 - Solver Configurations
- Results
 - Datasets
 - Derived Values
 - Surface Average 1
- Tables
- Velocity (spf)
 - Volume 1
- Pressure (spf)
- Concentration, Streamline (tds)
- Concentration, Surface (tds)
- Export
- Reports

Settings

Surface Average

[-] Evaluate

Label: Surface Average 1

Dataset: Study 2/Solution 2 (sol2)

Time selection: All

Selection: Manual

3

Expression	Unit	Description
c	mol/m ³	Molar concentration, c

Expression: c

Description: Molar concentration, c

Integration Settings

Graphics

Convergence Plot 1 × Function Plot ×

→ 3 | Up

3D model of a cylindrical component with a green circular surface highlighted on one of its ends, representing the outflow surface (Side 3). The model is shown in a perspective view with coordinate axes (x, y, z) and dimensions in meters (m).

- Select Study 2 (Time-dependent)
- Select the Outflow surface (Side 3)
- Input “c” as the expression to average
- Click on Evaluate

Model Builder

Type filter text

H2O (water) [liquid] (mat1)

NaOH [liquid] (mat2)

Basic (def)

Thermal expansion (ThermalE)

Piecewise (dL)

Piecewise 1 (CTE)

Laminar Flow (spf)

Fluid Properties 1

Initial Values 1

Wall 1

Inlet 1

Outlet 1

Transport of Diluted Species (tds)

Fluid 1

No Flux 1

Initial Values 1

Inflow 1

Outflow 1

Multiphysics

Reacting Flow, Diluted Species 1 (

Mesh 1

Study 1

Step 1: Stationary

Solver Configurations

Study 2

Step 1: Time Dependent

Solver Configurations

Results

Datasets

Derived Values

Surface Average 1

Tables

Velocity (spf)

Volume 1

Pressure (spf)

Concentration, Streamline (tds)

Concentration, Surface (tds)

1D Plot Group 5

Table Graph 1

Export

Settings

Table Graph

Plot

Label: Table Graph 1

Data

Source: Table

Table: Table 1

Row-based

x-axis data: Automatic

Plot columns: All excluding x-axis

Transformation: None

Sorting: None

Plot imaginary part

Preprocessing

Coloring and Style

Line style

Line: Solid

Color: Cycle

Color cycle: Default

Width: Default from preferences (1)

Line markers

Marker: None

Legends

Graphics

Convergence Plot 1

Function Plot

Molar concentration, c (mol/m³)

Time (s)

Messages

Progress

Log

Table 1

8.85 e-12

AUTO

8.5 e-1

850 e-3

0.85

Time (s)

Molar concentration, c (mol/m³)

0.0000

-1.1620E-18

0.10000

2.5788

0.20000

5.8548

0.30000

7.7052

0.40000

8.7292

Tutorial

Artificial Intelligence in Microfluidic Circuits



Speakers

Jorge Torres Gomez, Murat Kuşçu, Cansu Canbek, Volkan Can

ACM NanoCom 2026

Memorial University

Canada

September 21-23, 2026

